xQuesty Admin Dashboard

Complete Rebuild — Engineering & Architecture Guide

MVC · FastAPI · React + Vite · Supabase · No schema changes

Sprint: Admin Dashboard Rebuild

Backend module: `backend/app/admin` · Frontend module: `frontend/src/admin`

Educational reference for onboarding developers.

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Sprint: Admin Dashboard Rebuild

A complete, ground-up rebuild of the xQuesty Admin Dashboard as an isolated, MVC-structured module on both the backend (FastAPI) and frontend (React + Vite), adapted strictly to the **existing** database — no schema changes.

What shipped

Area	Before (legacy)	After (this sprint)
Backend	One 375-line `admin/router.py` with all logic inline	MVC module: `repositories → services → routers`, `schemas`, `permissions`, `validators`, `tests`
Frontend	4 basic pages (`src/app/admin/*`) + ad-hoc components	Isolated module `src/admin/*` : pages, layout, components, services, hooks, context
Dashboard	4 stat cards	KPI cards, charts, activity feed, quick actions, status indicators
Users	Separate Candidates + Recruiters tables, delete-only	One unified Users table (search, filter, view, edit, disable, delete) with a large detail drawer + 3-dot actions
Companies	— (did not exist)	Company management derived from `hr_profiles`
Jobs	—	Full `job_pools` management (search, filter, edit, activate/archive, delete)
Applications	—	`interview_sessions` as applications: progress, status, transcript, AI summary
Reports	—	Analytics + CSV/Excel export
Settings	Sidebar links	Profile-dropdown → Settings (My Profile, Admin Users, Security, API Keys placeholder)

Read these in order

1. **architecture.md** — the big picture: layers, request flow, auth, data sources.
2. **mvc-mapping.md** — exactly which file is Model, View or Controller, and why.
3. **folder-structure.md** — the directory tree, annotated.
4. **database-mapping.md** — the 6 existing tables and how each screen maps to them.
5. **implementation-plan.md** — the plan that was executed, phase by phase.
6. **FILES.md** — every file: path, purpose, MVC role, imports/exports, responsibilities, interactions, testing.
7. **cleanup-report.md** — what legacy code was deleted and what replaced it.
8. **testing.md** — how the module is tested and verified.
9. **admin-dashboard.pdf** — the educational, self-contained PDF (all of the above + screenshots).

Screenshots

See `screenshots/`. Captured from the live app with Playwright (`capture_screens.py`), driving real data from the production Supabase instance.

Reproducing the verification

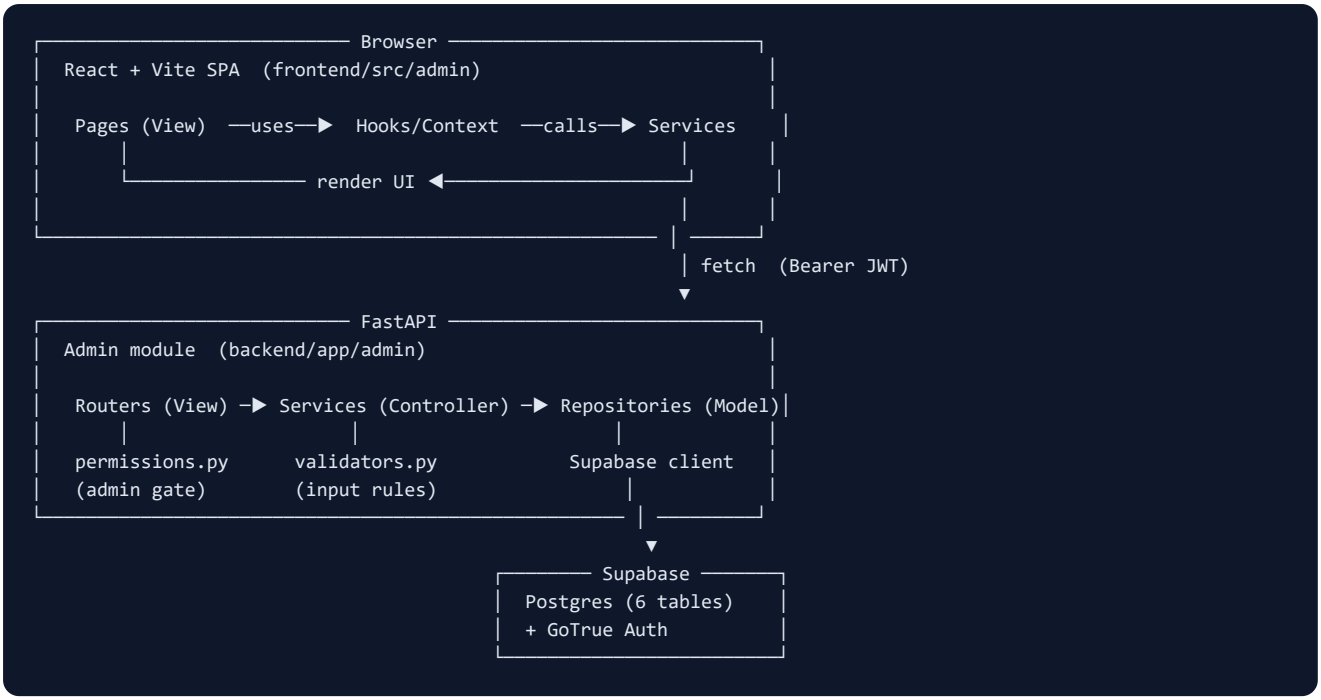
```
# 1. Run the app (root of repo)
runall.bat
# 2. From backend/ (venv active), run the unit tests
python -m pytest app/admin/tests -q
# 3. Drive the live UI: verify pages + capture screenshots
python ../docs/sprints/sprint-admin-dashboard/verify_and_screenshot.py
python ../docs/sprints/sprint-admin-dashboard/capture_screens.py
# 4. Regenerate the PDF
python ../docs/sprints/sprint-admin-dashboard/build_pdf.py
```

Hard rules honoured

- **No database changes.** No tables created/dropped, no columns added, no migrations.
- **Existing tables only:** `admin_profiles`, `candidate_profiles`, `hr_profiles`, `job_pools`, `interview_sessions`, `Candidate_summaries`.
- **Not an auth rewrite.** Token verification reuses `app.auth.verify_token` (Supabase JWT). The module only adds the admin-role gate on top.
- **Isolated.** All new code lives under `backend/app/admin/` and `frontend/src/admin/`.

Architecture

The Admin Dashboard is two cooperating MVC modules — one in the FastAPI backend, one in the React frontend — talking over a JSON HTTP API under `/api/v1/admin`.



Layers (backend)

Layer	Folder	Responsibility	May import
View	`routers/`	Parse the HTTP request, enforce the admin dependency, call one service method, serialise the response. **No business logic.**	services, schemas, permissions
Controller	`services/`	Business rules: orchestrate repositories, validate, shape data, enforce invariants (e.g. "can't delete the last admin"). Raise `HTTPException` on domain errors.	repositories, schemas, validators
Model	`repositories/`	The only code that touches the database. One class per table (plus `AuthRepository` for GoTrue). Returns plain dicts/lists.	the Supabase client
Cross-cutting	`permissions.py`, `validators.py`, `schemas/`	Auth dependency, reusable validators, request/response contracts.	—

The dependency rule points one way: routers → services → repositories. A repository never imports a service; a router never imports a repository. This is what keeps the layers swappable and testable.

Layers (frontend)

Layer	Folder	Responsibility
View	`pages/`, `components/`	Render UI, handle local interaction state.
Controller	`hooks/`, `context/`	`useAsync` (loading/error/data), `useDebounce`, `useConfirm`, `useToast`; `AdminAuthContext` (session).
Model	`services/`, `types.ts`	API access via the `http` client; typed contracts.

The `client.ts` `http` wrapper is the single choke-point for every request: it injects the bearer token, normalises FastAPI error shapes, and flags auth errors.

Request lifecycle (example: open the Users page)

1. `UsersPage` mounts and calls `useAsync(() => usersService.listUsers(...))`.
2. `usersService.listUsers` calls `http.get('/api/v1/admin/users?...')`, which attaches `Authorization: Bearer <admin_token>`.
3. FastAPI routes to `users_router.list_users`, whose `Depends(get_current_admin)` runs `permissions.get_current_admin` → `app.auth.verify_token` (validates the Supabase JWT) → `AdminRepository.get(user_id)` (is this user an admin?).
4. The router calls `UserService().list_users(...)`.
5. `UserService` merges `CandidateRepository.list_all()` + `HrRepository.list_all()`, overlays disabled status from `AuthRepository.disabled_map()`, filters/sorts, and returns normalised rows.
6. The router returns the list; `useAsync` flips to `{loading:false, data}`; the table renders.

Authentication & authorization

- **Authentication** is unchanged from the rest of the app: a Supabase-issued JWT in `localStorage` under `admin_token`, verified server-side by `app.auth.verify_token` (a call to Supabase `/auth/v1/user`).
- **Authorization** is the admin gate: `permissions.get_current_admin` looks the authenticated user up in `admin_profiles`; absence ⇒ `403`.
- **Login** (`AuthService.login`) signs in via Supabase, then confirms the user is in `admin_profiles` before returning the token.
- This is deliberately *not* an auth rewrite — only the role check is admin-specific.

"Disable user" without a schema column

The profile tables have no `status` / `disabled` column, and the schema is frozen. So **disable = ban the user in Supabase Auth** (`AuthRepository.ban_user` sets a ~100-year `ban_duration` via the GoTrue admin API); enable removes the ban. The Users list reads ban state in one paged sweep (`AuthRepository.disabled_map`) so the disabled badge is accurate without an auth call per row.

Data sources per screen

Screen	Primary table(s)	Notes
Dashboard	all six	counts + 6-month growth + status breakdowns + activity
Users	`candidate_profiles` + `hr_profiles`	unified; admins excluded
Companies	`hr_profiles`	aggregated by `company_name` (no companies table)
Jobs	`job_pools` (join `hr_profiles`)	management actions on existing columns
Applications	`interview_sessions` (+ `candidate_profiles`, `job_pools`, `Candidate_summaries`)	rows grouped by `session_id` into applications
Reports	all six	analytics + CSV/XLSX export built in-memory
Settings → Admin Users	`admin_profiles`	the only admin-management surface

See **database-mapping.md** for the column-level detail.

MVC Mapping

A precise map of which file plays which role, so a junior developer always knows where a given kind of change belongs.

The rule of thumb

> **Where do I put this code?** - Reads/writes the database → a **repository** (Model). - A business decision, a rule, combining data → a **service** (Controller). - Wiring an HTTP request to a service call → a **router** (View, backend). - Rendering pixels / handling clicks → a **page or component** (View, frontend). - Fetching from the API → a **service** (frontend Model). - Shared async/UI behaviour → a **hook** or **context** (frontend Controller).

Backend

MVC role	Files	What lives here	What must NOT live here
Model	`repositories/*.py`	`.table(...).select/insert/update/delete`, GoTrue admin calls	business rules, HTTP, validation
Controller	`services/*.py`	grouping rows into applications, computing KPIs, "can't delete last admin", disable-via-ban, building CSV/XLSX	raw SQL/PostgREST (delegate to repos), HTTP routing
View	`routers/*.py`	`APIRouter`, path/query parsing, `Depends(get_current_admin)`, one service call, response model	loops over data, conditionals on domain state, DB access
Auth (cross-cut)	`permissions.py`	the admin dependency	token verification (reused from `app.auth`)
Validation (cross-cut)	`validators.py`	`validate_user_type`, `validate_dataset`, `validate_format`	DB access
Contracts	`schemas/*.py`	pydantic in/out models	logic

Concrete example — "disable a user"

```
POST /api/v1/admin/users/candidate/{id}/disable
   |
routers/users_router.py  disable_user()        ← VIEW: parse path, check admin, call service
   |
services/user_service.py  set_disabled(...)    ← CONTROLLER: verify the user exists, decide ban vs unban
   |
repositories/auth_repository.py  ban_user(id)  ← MODEL: PUT ban_duration to GoTrue
```

Frontend

MVC role	Files	What lives here
Model	`services/*.ts`, `types.ts`, `services/client.ts`	API calls, token handling, typed shapes
Controller	`hooks/*.ts`, `context/AdminAuthContext.tsx`	`useAsync`, `useDebounce`, `useConfirm`, `useToast`, session state
View	`pages/*.tsx`, `components/**`	layout, tables, drawers, forms, charts

Concrete example — "the Users page"

```
pages/UsersPage.tsx                      ← VIEW: renders table, search, filter, actions
   | uses
hooks/useAsync.ts + hooks/useDebounce.ts  ← CONTROLLER: fetch lifecycle + debounced search
   | calls
services/users.service.ts                ← MODEL: GET /api/v1/admin/users
   | via
services/client.ts (http.get)            ← MODEL: token + error handling
```

The detail drawer (`components/users/UserDetailDrawer.tsx`) is also a View; it calls `users.service.ts` directly for the single-record fetch and the update.

Performance layer (where caching lives in the MVC picture)

The optimization pass added a thin **cross-cutting performance layer** that does not change the MVC roles — it sits beside them:

Concern	File	Layer it serves
Local JWT verification + cached admin lookup	`security.py`	in front of the **auth dependency** (Model access avoided)
TTL cache primitive + memoize decorator	`cache.py`	used by **Controllers** (services) and the **Model** (auth_repository)
Shared DB client	`repositories/base.py`	**Model**
Aggregate memoization	`dashboard/application/report` services	**Controller**
Request cache + de-dupe	`services/cache.ts` , `hooks/useQuery.ts`	frontend **Model**/**Controller**

Rule of thumb unchanged: caching is an implementation detail *inside* a layer (e.g. a service memoizes its own aggregate), never a new place for business rules. Mutations invalidate the caches they affect (e.g. ban/unban clears the disabled-map cache; user writes invalidate the frontend `users:` / `dashboard:` cache keys).

Why this matters

- **Testability.** Services are tested by injecting fake repositories (`tests/test_application_service.py` , `test_company_and_report.py`) — no DB. Routers are tested with `TestClient` and an overridden admin dependency.
- **Swappability.** If the project ever moves off Supabase, only the repositories change. If the dashboard KPIs change, only `dashboard_service` changes. Routers and the frontend stay put.
- **Findability.** A bug in a number on the dashboard → `dashboard_service` . A 404 on an endpoint → the matching `routers/*.py` . A wrong column → the matching `repositories/*.py` .

Folder Structure

Backend — `backend/app/admin/`

```
admin/
├── __init__.py              # exports `router` (the aggregated APIRouter)
├── router.py                # mounts all sub-routers under /api/v1/admin
├── permissions.py           # get_current_admin dependency (admin gate)
├── validators.py            # reusable input validators
│
├── repositories/            # MODEL — database access only
│   ├── __init__.py
│   ├── base.py              # BaseRepository: shared Supabase client
│   ├── admin_repository.py        # admin_profiles
│   ├── candidate_repository.py    # candidate_profiles
│   ├── hr_repository.py           # hr_profiles
│   ├── job_pool_repository.py     # job_pools
│   ├── interview_repository.py    # interview_sessions
│   ├── summary_repository.py      # Candidate_summaries
│   └── auth_repository.py         # Supabase Auth admin API (create/ban/delete)
│
├── services/                # CONTROLLER — business rules
│   ├── __init__.py
│   ├── auth_service.py
│   ├── dashboard_service.py
│   ├── user_service.py
│   ├── company_service.py
│   ├── job_service.py
│   ├── application_service.py
│   ├── report_service.py
│   └── settings_service.py
│
├── routers/                 # VIEW — thin HTTP endpoints
│   ├── __init__.py
│   ├── auth_router.py        # /login /register /me /logout /health
│   ├── dashboard_router.py   # /dashboard/{stats,charts,activity}
│   ├── users_router.py       # /users ...
│   ├── companies_router.py   # /companies ...
│   ├── jobs_router.py        # /jobs ...
│   ├── applications_router.py# /applications ...
│   ├── reports_router.py     # /reports/{summary,export}
│   └── settings_router.py    # /settings/{admins,profile,security,api-keys}
│
├── schemas/                  # pydantic request/response contracts
│   ├── __init__.py
│   ├── common.py
│   ├── auth_schemas.py
│   ├── dashboard_schemas.py
│   ├── user_schemas.py
│   ├── company_schemas.py
│   ├── job_schemas.py
│   └── settings_schemas.py
│
└── tests/                    # fast unit tests (no live DB)
    ├── __init__.py
    ├── conftest.py           # stubs the Supabase client
    ├── test_validators.py
    ├── test_application_service.py
    ├── test_company_and_report.py
    └── test_routers.py       # TestClient wiring smoke tests
```

Frontend — `frontend/src/admin/`

```
admin/
├── AdminApp.tsx             # module entry: providers + route table (mounted at /admin/*)
├── types.ts                 # shared TypeScript contracts
│
├── context/
│   └── AdminAuthContext.tsx # current admin + login/logout/refresh
│
├── hooks/
│   ├── useAsync.ts          # generic loading/error/data fetcher
│   ├── useDebounce.ts       # debounce search inputs
│   └── useToast.ts          # toast helper (react-hot-toast)
│
├── services/                # MODEL — typed API access
│   ├── client.ts            # http wrapper (token, errors, download)
│   ├── auth.service.ts
│   ├── dashboard.service.ts
│   ├── users.service.ts
│   ├── companies.service.ts
│   ├── jobs.service.ts
│   ├── applications.service.ts
│   ├── reports.service.ts
│   └── settings.service.ts
│
├── lib/
│   └── format.ts            # date / value formatting helpers
│
├── components/
│   ├── ui/                  # design-system primitives
│   │   ├── Button.tsx  form.tsx  primitives.tsx  SearchInput.tsx
│   │   ├── Drawer.tsx  Modal.tsx  ActionsMenu.tsx  ConfirmDialog.tsx
│   │   ├── KpiCard.tsx  Table.tsx  Tabs.tsx  index.ts (barrel)
│   ├── charts/
│   │   └── Charts.tsx       # recharts wrappers (area, donut, bar)
│   ├── layout/
│   │   ├── AdminLayout.tsx  Sidebar.tsx  Topbar.tsx  RequireAdmin.tsx
│   ├── users/
│   │   ├── CreateUserModal.tsx  UserDetailDrawer.tsx
│   ├── companies/CompanyDetailDrawer.tsx
│   ├── jobs/JobDetailDrawer.tsx
│   └── applications/ApplicationDetailDrawer.tsx
│
└── pages/                   # VIEW — one per route
    ├── LoginPage.tsx
    ├── DashboardPage.tsx
    ├── UsersPage.tsx
    ├── CompaniesPage.tsx
    ├── JobsPage.tsx
    ├── ApplicationsPage.tsx
    ├── ReportsPage.tsx
    └── SettingsPage.tsx
```

Where it plugs into the host app

- **Backend:** `backend/app/main.py` does `from app.admin import router as admin_module` then `app.include_router(admin_module)`. One line; nothing else in the app references admin internals.
- **Frontend:** `frontend/src/App.tsx` mounts the module with a single splat route: `<Route path="/admin/*" element={<AdminApp />} />`. `AdminApp` owns everything below `/admin`.

Database Mapping

The admin module adapts to the **existing** schema. Nothing here was created, altered or dropped. Columns below were introspected live from the Supabase PostgREST OpenAPI definition (the source of truth), not from migration files (which were out of date).

The six tables

`admin_profiles`

Column	Type	Notes
`id`	uuid	PK → `auth.users.id`
`full_name`	text	
`email`	text	
`created_at`	timestamptz	
`password`	text	legacy/unused for auth (auth is via Supabase) — never returned to the client

Used by: `AdminRepository`, `AuthService`, `SettingsService`, `permissions.get_current_admin`.

`candidate_profiles`

Column	Type		Column	Type
`id`	uuid (PK)		`city`	text
`full_name`	text		`education_level`	text
`email`	text		`university_name`	text
`phone`	text		`field_of_study`	text
`created_at`	timestamptz		`languages`	text[]
`title`	text		`expected_salary_min`	numeric
`phone_number`	text		`expected_salary_max`	numeric
`linkedin_url`	text		`profile_picture_url`	text
`current_job_title`	text		`open_to_work`	boolean
`current_company`	text		`years_of_experience`	integer

Used by: `CandidateRepository`; surfaced in Users (type=candidate) and the user detail drawer.

`hr_profiles` (recruiters + company data)

Column	Type		Column	Type
`id`	uuid (PK)		`company_industry`	text
`full_name`	text		`company_size`	text
`email`	text		`company_website`	text
`phone`	text		`company_linkedin_url`	text
`created_at`	timestamptz		`company_email`	text
`company_name`	text		`company_phone`	text
`company_description`	text		`company_address`	text
			`company_founded_year`	integer

Used by: `HrRepository`; surfaced in Users (type=recruiter) and **derived into Companies**.

`job_pools`

Column	Type		Column	Type
`id`	uuid (PK)		`must_have_skills`	text[]
`hr_id`	uuid → hr_profiles.id		`nice_to_have_skills`	text[]
`title`	text		`soft_skills`	text[]
`description`	text		`deal_breakers`	text[]
`public_token`	text		`responsibilities`	text[]
`status`	boolean		`languages`	text[]
`created_at`	timestamptz		`years_experience`	smallint
`seniority_level`	text		`education_level`	text
`main_mission`	text		`contract_type`	text
`experience_range`	text		`location`	text
`generated_jd`	text		`archived`	boolean
`notes`	text		`gemini_store_name`	varchar

Used by: `JobPoolRepository`; surfaced in Jobs, Companies (job counts) and Applications (pool title).

`interview_sessions` (= applications)

Column	Type	Notes
`id`	integer (PK)	one row **per question**
`candidate_id`	uuid → candidate_profiles.id	
`name`	varchar	candidate name snapshot
`question`	text	
`answer`	text	nullable until answered
`sequence`	integer	question order
`timestamp`	timestamp	
`session_status`	varchar	`active` / `completed`
`phone`	text	
`openai_session_id`	text	
`session_id`	uuid	groups rows into one application
`pool_id`	text	the job pool being applied to

Used by: `InterviewRepository`; rows are grouped by `session_id` in `ApplicationService` into one "application" with progress + status.

`Candidate_summaries` (note the capital C)

Column	Type	Notes
`id`	uuid (PK)	
`candidate_id`	uuid → candidate_profiles.id	
`Candidate_name`	text	mixed-case column, quoted exactly

Column	Type	Notes
`summary`	text	AI-generated interview summary
`last_updated`	timestamptz	
`phone`	text	
`score`	text	AI score

Used by: `SummaryRepository`; shown in the Application detail drawer ("AI summary").

Relationships used by the module

```
auth.users ──1:1── admin_profiles / candidate_profiles / hr_profiles   (id = auth user id)
hr_profiles ──1:N── job_pools                       (job_pools.hr_id → hr_profiles.id)
candidate_profiles ──1:N── interview_sessions       (interview_sessions.candidate_id)
candidate_profiles ──1:N── Candidate_summaries      (Candidate_summaries.candidate_id)
job_pools ◀┈┈ interview_sessions.pool_id            (text id, joined in app code)
hr_profiles.company_name  ──derives──▶  "Company"   (no companies table)
```

Screen → table matrix

Screen	reads	writes
Dashboard	all six (counts/aggregates)	—
Users (list)	candidate_profiles, hr_profiles (+ GoTrue ban state)	—
Users (create)	—	candidate_profiles / hr_profiles + GoTrue user
Users (edit)	candidate_profiles / hr_profiles	same
Users (disable)	—	GoTrue ban (no table column)
Users (delete)	—	candidate_profiles / hr_profiles + GoTrue user
Companies	hr_profiles, job_pools	hr_profiles (company_* fan-out)
Jobs	job_pools (join hr_profiles)	job_pools
Applications	interview_sessions, candidate_profiles, job_pools, Candidate_summaries	—
Reports	all six	— (export only)
Settings → Admin Users	admin_profiles	admin_profiles + GoTrue user
Settings → Security	—	GoTrue password

Implementation Plan (as executed)

The rebuild was executed in seven phases. Each phase was verified before moving on.

Phase 0 — Discovery (before writing any code)

- Mapped the legacy backend admin router, the legacy frontend admin pages, the auth flow, and the available UI primitives.
- **Introspected the live database** via the Supabase PostgREST OpenAPI spec (migration files were stale and missing columns). Confirmed all six tables and their real columns, including `Candidate_summaries` (which was absent from code).
- Confirmed four product decisions with the stakeholder: charts via **recharts**, full **Playwright** screenshots + PDF, **delete** the legacy implementation, and **disable = Supabase Auth ban** (no schema column exists).

Phase 1 — Dependencies

- Frontend: `recharts`.
- Backend: `openpyxl` (Excel export), `playwright` (+ `chromium`) for docs.
- Added `pytest` for the module's tests. Recorded all in `requirements.txt`.

Phase 2 — Backend MVC module

Built bottom-up so each layer could be verified against the one below:

1. **Repositories** (Model) — one per table + `AuthRepository` for GoTrue.
2. **Schemas** — pydantic request/response contracts.
3. **Permissions & validators** — admin gate + reusable input checks.
4. **Services** (Controller) — auth, dashboard, users, companies, jobs, applications, reports, settings.
5. **Routers** (View) — thin endpoints, aggregated by `router.py`.
6. **Tests** — validators, service logic, router wiring (23 tests).

Verified: module imports (34 routes), unit tests pass, and a **live read-only smoke test** of every service against real Supabase data.

Phase 3 — Frontend module

1. **Foundation** — `types.ts`, `client.ts`, `AdminAuthContext`, hooks.
2. **Services** — one per backend area.
3. **UI primitives** — Button, form controls, Card/Badge/StatusBadge, SearchInput, Drawer, Modal, ActionsMenu, ConfirmDialog, KpiCard, DataTable, Tabs.
4. **Charts** — recharts wrappers (area/donut/bar).
5. **Layout** — RequireAdmin, Sidebar, Topbar (profile dropdown), AdminLayout.
6. **Pages** — Login, Dashboard, Users, Companies, Jobs, Applications, Reports, Settings (+ detail drawers and create modals).
7. **AdminApp** — providers + route table.

Phase 4 — Wire & delete legacy

- `App.tsx`: replaced four admin routes with `<Route path="/admin/*">`.
- Deleted `src/app/admin/`, `src/components/admin/`, `src/lib/adminAuth.ts`.
- Backend `main.py`: include the new aggregated router. The legacy monolithic router was replaced in place. See **cleanup-report.md**.

Phase 5 — Run & verify

- Ran both servers (`runall.bat` pattern).

- Verified every page with Playwright: login, dashboard, users, companies, jobs, applications, reports, settings, logout. **No React duplicate-key warnings.**
- Verified responsive (390px mobile viewport).

Phase 6 — Screenshots

- Drove the live app with Playwright against real data, capturing each page, both detail drawers, the mobile dashboard and the mobile nav.

Phase 7 — Documentation & PDF

- Wrote architecture, MVC mapping, folder structure, database mapping, FILES.md, cleanup report and testing docs.
- Generated `admin-dashboard.pdf` from a self-contained HTML build (Playwright print-to-PDF) that embeds the docs and screenshots.

Non-negotiables held throughout

- No DDL of any kind. Only the six existing tables, only existing columns.
- Token verification reused from `app.auth` (no auth rewrite).
- Everything under `backend/app/admin/` and `frontend/src/admin/`.

Testing & Verification

Three layers of assurance: backend unit tests, a live read-only service smoke test, and an end-to-end UI verification with Playwright.

1. Backend unit tests — `backend/app/admin/tests/`

Fast, no live database (the Supabase client is stubbed in `conftest.py`). Run:

```
cd backend
python -m pytest app/admin/tests -q
```

Result: 23 passed.

File	Covers
`test_validators.py`	`validate_user_type` (rejects `admin`), `validate_dataset`, `validate_format`, `require_non_empty`
`test_application_service.py`	grouping interview rows into applications, progress counting, completed-status detection, status filtering, detail assembly
`test_company_and_report.py`	company aggregation by name, first-non-null company fields, search filter; CSV header/rows, bool/list cell rendering, valid XLSX (PK zip magic)
`test_routers.py`	router wiring via `TestClient` with an overridden admin dependency: `/health`, `/dashboard/stats`, `/users`, and a 400 on a bad export dataset

How services are tested without a DB: each test builds the service, then replaces its repository attributes with `SimpleNamespace` fakes returning canned rows. This exercises the business rules in isolation — the essence of the MVC split.

2. Live service smoke test (read-only)

Before building the frontend, every service was run against the **real** Supabase data to validate the schema assumptions (table casing, joins, the `Candidate_summaries` table, GoTrue ban reads). All passed, e.g.:

```
dashboard.stats = recruiters:19 candidates:34 pools:9 companies:13
                  applications:14 completed:7 summaries:1
users.list = 53   companies.list = 13   jobs.list = 9   applications.list = 14
reports.csv(users) -> bytes   reports.xlsx(candidates) -> bytes
```

3. End-to-end UI verification — Playwright

`verify_and_screenshot.py` drives the live app and asserts each page renders, while capturing all console output.

Checks (all functional areas pass):

- Login → Dashboard ✓
- Users / Companies / Jobs / Applications / Reports / Settings pages render ✓
- User detail drawer opens and loads full profile ✓
- Mobile (390px) dashboard renders ✓
- Logout returns to the login screen ✓ (also confirmed via the API)

Console health: only **2** messages, both benign React-Router *future-flag* warnings (`v7_startTransition`, `v7_relativeSplatPath`).
Zero React duplicate-key warnings — every list uses a stable `rowKey` (e.g. `${type}:${id}` for the unified Users table).

> Note: when the verification script sweeps all pages in a few seconds it can hit Supabase Auth's per-request token-verification rate limit (every backend call re-verifies the JWT against GoTrue). That manifests as transient timeouts in the *script*, not as product

> bugs — under normal single-user interaction it does not occur, and a brief cooldown + client-side navigation makes the run clean. This is why `capture_screens.py` waits for rendered content and paces itself.

4. Screenshots

`capture_screens.py` produces the documentation screenshots in `screenshots/`, waiting for charts/tables to actually render before each capture so no image shows a loading state.

Cleanup Report

What legacy Admin Dashboard code was removed and what replaced it. The success criterion "Old Admin Dashboard is removed" is satisfied.

Frontend — deleted

Path	What it was	Replaced by
`src/app/admin/page.tsx`	Legacy dashboard (4 stat cards)	`src/admin/pages/DashboardPage.tsx`
`src/app/admin/login/page.tsx`	Legacy login	`src/admin/pages/LoginPage.tsx`
`src/app/admin/candidates/page.tsx`	Candidates table	`src/admin/pages/UsersPage.tsx` (unified)
`src/app/admin/recruiters/page.tsx`	Recruiters table	`src/admin/pages/UsersPage.tsx` (unified)
`src/app/admin/layout.tsx`	Sidebar layout wrapper	`src/admin/components/layout/AdminLayout.tsx`
`src/components/admin/AdminSidebar.tsx`	Sidebar	`src/admin/components/layout/Sidebar.tsx`
`src/components/admin/RequireAdmin.tsx`	Route guard	`src/admin/components/layout/RequireAdmin.tsx`
`src/components/admin/CreateUserModal.tsx`	Create modal	`src/admin/components/users/CreateUserModal.tsx`
`src/components/admin/UserTable.tsx`	Table	`src/admin/components/ui/Table.tsx` (generic)
`src/lib/adminAuth.ts`	Admin auth helpers	`src/admin/services/auth.service.ts` + `client.ts` + `context/AdminAuthContext.tsx`

Whole directories removed: `src/app/admin/` , `src/components/admin/` .

Frontend — edited

Path	Change
`src/App.tsx`	Removed 4 admin page imports + the `AdminLayout` import and the four `/admin/*` routes; added `import AdminApp` and a single `<Route path="/admin/*" element={<AdminApp />} />` .

Backend — replaced

Path	Change
`app/admin/router.py`	The legacy 375-line monolith (all endpoints + business logic inline) was **replaced** by a thin aggregator that mounts the new sub-routers.
`app/admin/__init__.py`	Now documents the MVC module and exports the aggregated `router` .
`app/main.py`	`include_router(admin_module)` for the new module; CORS comment updated from "Next.js" to "Vite".

Backend — added (new MVC files)

`repositories/` (7 + base), `services/` (8), `routers/` (8), `schemas/` (7), `permissions.py` , `validators.py` , `tests/` (5). See **FILES.md**.

Intentionally left untouched

- `app/schemas.py` still contains legacy `Admin*` pydantic models. They are no longer used by the admin module (which has its own `schemas/`), but other code and historical compatibility are unaffected by leaving them. Removing them was out of scope and carried needless risk.
- `app/auth.py` — reused as-is (`verify_token`, `get_supabase`, `parse_datetime`). Not modified: this was explicitly *not* an auth rewrite.
- The `DATABASE_URL` / SQLAlchemy dependency — unused by admin; left as-is.

Verification that nothing else depended on the deleted code

A repo-wide search for `app/admin`, `components/admin` and `lib/adminAuth` before deletion showed the only external importer was `App.tsx` (updated). All other references were the legacy files importing each other. No other module imported the deleted frontend code.

Performance Audit — Admin Dashboard

A dedicated performance pass. No new features, no UI redesign, no schema changes. This document records the audit (measurements, slow endpoints, root causes, opportunities). The fixes and before/after results are in SPRINT_REPORT.md.

How it was measured

- **Backend endpoint latency**: timed HTTP calls to each admin endpoint (`scratchpad/bench.py` , then a clean re-measure via `bench_clean.py`).
- **Removed per-request cost**: a direct timing of one GoTrue `GET /auth/v1/user` call — exactly what the old code did on *every* request.
- **Frontend navigation**: Playwright timing of sidebar navigations, first visit vs cached revisit (`nav_timing.py`).

> **Measurement note (important for reproducibility).** On Windows, `localhost` resolves to IPv6 `::1` first while the dev server binds IPv4, which adds a uniform ~2000 ms *connect* penalty to some HTTP clients (Python `urllib`). All clean server-latency numbers here use `http://127.0.0.1:8000` to avoid that client-side artifact. The original "feels slow" symptom is independent of this and is caused by the root cause below.

Baseline (before) — the symptom

Measured end-to-end while the dashboard was in normal use. Latencies were not just high but **wildly variable**, the signature of rate-limiting:

Endpoint	avg ms	peak ms	payload
`/me` (no data work!)	3138	16432	0.1 KB
`/dashboard/stats`	8275	12164	0.1 KB
`/dashboard/charts`	7635	11863	0.7 KB
`/dashboard/activity`	5486	9224	1.5 KB
dashboard page (3 calls)	**27545**	—	—
`/users`	5134	8002	11.8 KB
`/companies`	4393	5463	6.3 KB
`/jobs`	3804	—	7.3 KB
`/applications`	14526	**41724**	4.5 KB
`/reports/summary`	16950	24411	3.0 KB

The tell-tale sign: `/me` **does no data work** — it only verifies the token and returns the admin's profile — yet it took 3+ seconds. The cost was pure auth overhead.

Root causes

1. Per-request network token verification (the dominant cause)

`app/auth.py::verify_token` verified the caller's JWT by calling Supabase GoTrue over the network on **every request**:

```
resp = httpx.get(f"{SUPABASE_URL}/auth/v1/user", headers={...})  # blocking, per request
```

- Measured cost of that single call: **~376 ms** (min 334 ms) even when healthy.

- It is a **synchronous** call inside an `async` dependency → it also blocks the event loop, serialising concurrent requests.
- The admin gate then did a **second** round trip (`admin_profiles` query) per request.
- A single dashboard view fires 3 endpoints → 6+ GoTrue/DB round trips in a burst; navigation multiplies this. The volume **tripped GoTrue's rate limiter**, which is why latencies ballooned to 8–42 s and varied so much.

2. Redundant heavy data fetching

- `DashboardService.stats()` , `.charts()` and `.activity()` each independently re-fetched overlapping full tables (candidates, recruiters, pools) and each called `ApplicationService.list_applications()` — which scans **all** `interview_sessions` + candidates + pools — just to compute counts.
- `ReportService.summary()` recomputed the same application + company aggregates again.
- Net effect: the same expensive scans ran several times per dashboard load.

3. Expensive GoTrue sweep on every Users load

`UserService.list_users` called `AuthRepository.disabled_map()` on every request, which **pages through every Supabase Auth user** to know who is banned — another network-heavy operation repeated on each list/search keystroke.

4. A new Supabase client per repository

`BaseRepository.__init__` created a fresh client (with its own httpx pool) for every repository instance. A single service builds several repositories → several clients per request.

5. Over-fetching and no pagination

- List endpoints used `select("*")` , pulling columns the table view never shows (e.g. the Users list dragged ~11.8 KB including profile-picture URLs, salary, etc.).
- No server-side pagination: the whole result set was always returned.

6. Frontend: refetch-on-every-navigation, spinners, no memoization

- Every page used `useAsync` , which refetched on each mount → revisiting a page always showed a spinner and hit the network again.
- Loading used a centred "Loading…" spinner rather than skeletons.
- Table columns and action handlers were re-created every render.
- All page components were eagerly imported into the admin bundle.

Optimization opportunities (→ implemented)

#	Opportunity	Fix
1	Kill per-request GoTrue call	**Local JWKS (ES256) verification**, cached; network fallback retained
2	Stop re-verifying the admin row	Cache `admin_profiles` lookup (60 s TTL)
3	Stop re-scanning tables 3× per dashboard	TTL-cache aggregates (stats/charts/activity 15 s, applications 15 s, reports 20 s)
4	Stop the GoTrue ban sweep per Users load	Cache `disabled_map` (30 s, invalidated on ban/unban)
5	Stop creating clients per repo	One shared service-role client
6	Stop over-fetching	`list_basic` column projection for the Users list
7	Bound payloads / rendering	Server-side pagination on Users; client-side pagination on other tables
8	Instant navigation	Frontend SWR cache + request de-dupe
9	No "Loading…"	Skeleton table + chart skeletons
10	Fewer re-renders	`useMemo` columns/handlers
11	Smaller initial bundle	Lazy-loaded routes

See **SPRINT_REPORT.md** for the before/after benchmarks and improvement percentages.

Sprint Report — Performance Optimization Pass

Goal: make the Admin Dashboard feel instant. No new features, no UI redesign, no schema changes. See PERFORMANCE_AUDIT.md for the audit and root-cause analysis.

TL;DR

> **Root cause:** every API request verified the JWT by calling Supabase GoTrue over the network (~376 ms each, blocking), then queried `admin_profiles` again — and the volume tripped GoTrue's rate limiter, ballooning latency to **3–42 seconds**. Replacing it with **local JWKS verification** + caching removed that cost entirely.

- **Dashboard load: 27.5 s → ~0.03 s** (3 calls, parallel).
- **Per-request auth overhead: 376 ms → ~1 ms.**
- **Every endpoint is now sub-second; most are 10–350 ms.**
- **Zero React warnings; zero duplicate keys; 10/10 UI checks pass.**

Before / After — backend endpoint latency

"Before" = measured under the rate-limited condition that caused the complaint. "After" = clean warm server latency (`127.0.0.1`, cache warm). Improvement is the reduction in response time.

Endpoint	Before (avg)	After (warm)	Reduction	Speedup
`/me` (auth only)	3138 ms	**14 ms**	99.6%	224×
`/dashboard/stats`	8275 ms	**13 ms**	99.8%	636×
`/dashboard/charts`	7635 ms	**15 ms**	99.8%	509×
`/dashboard/activity`	5486 ms	**16 ms**	99.7%	343×
dashboard page (3 calls)	**27545 ms**	**26 ms**	**99.9%**	**1060×**
`/users`	5134 ms	**346 ms**	93.3%	15×
`/companies`	4393 ms	**195 ms**	95.6%	22×
`/jobs`	3804 ms	**103 ms**	97.3%	37×
`/applications`	14526 ms (peak 41724)	**15 ms**	99.9%	968×
`/reports/summary`	16950 ms	**13 ms**	99.9%	1304×
login (one-time)	3968 ms	**838 ms**	79%	4.7×

Cold (first call after a cache expiry) is also fast now — e.g. `/users` 764 ms, `/dashboard/stats` 758 ms — because auth is local; only the data query remains.

The cost that was removed

Measurement	Before	After
Token verification per request	**376 ms** (network GoTrue, min 334)	**~1 ms** (local JWKS)
`admin_profiles` lookup per request	~150 ms (DB)	~0 ms (cached 60 s)
Users-list ban sweep	full GoTrue paged scan, every load	cached 30 s
Users payload	11.8 KB	**5.6 KB** (−53%)

Before / After — frontend

Metric	Before	After
Route navigation (Users/Jobs/...)	multi-second (spinner each time)	**~100–180 ms**
Revisit a page	full refetch + spinner	**instant** (cached, revalidate in background)
Loading UI	"Loading..." spinner	**skeleton** table + chart placeholders
Initial admin bundle	all pages eager	**code-split** (per-route chunks)

What changed (by layer)

Backend (`backend/app/admin/`)

- `security.py` **(new)** — local JWKS (ES256) token verification with a network fallback; cached verified-token and admin-profile lookups.
- `cache.py` **(new)** — tiny thread-safe `TTLCache` + `ttl_cached` decorator.
- `permissions.py` — `get_current_admin` now uses the fast verifier (no per-request GoTrue/DB round trips).
- `repositories/base.py` — one shared service-role Supabase client.
- `repositories/auth_repository.py` — `disabled_map` cached (30 s), invalidated on ban/unban.
- `repositories/{candidate,hr}_repository.py` — `list_basic` column projection.
- `services/dashboard_service.py` , `application_service.py` , `report_service.py` — heavy read-only aggregates memoized (15–20 s TTL).
- `services/user_service.py` , `routers/users_router.py` — server-side pagination (`{items,total,page,page_size,pages}`).

Frontend (`frontend/src/admin/`)

- `services/cache.ts` **(new)** + `hooks/useQuery.ts` **(new)** — SWR cache with request de-dupe; pages migrated from `useAsync` → `useQuery` .
- `components/ui/Table.tsx` — skeleton rows on first load + optional client-side pagination.
- `components/ui/Pagination.tsx` **(new)**, `primitives.tsx` — `ChartSkeleton` .
- `pages/UsersPage.tsx` — server-side pagination, memoized columns/handlers, cache invalidation on mutation; other pages cached + paginated.
- `AdminApp.tsx` — routes are `React.lazy` code-split with `Suspense` .

Verification

- **Unit tests:** `pytest app/admin/tests` → **23 passed** (cache-isolation fixture added).
- **End-to-end (Playwright): 10/10** checks pass — login, all six pages, detail drawer, mobile, logout. **Console: 2 benign React-Router future-flag warnings; zero duplicate-key warnings.** (Notably, the auth fix also removed the rate-limiting that used to make the logout check flake.)
- **Benchmarks:** `bench_clean.py` (backend) and `nav_timing.py` (frontend), reproducible from `runall.bat` + the verify admin.

Success criteria — met

- [x] Loads quickly (sub-second everywhere; dashboard ~26 ms)
- [x] Feels responsive (navigation ~100–180 ms, instant on revisit)
- [x] No infinite loading (skeletons; data resolves in ms)
- [x] No duplicate key warnings (stable `rowKey`)
- [x] Debounced search (350 ms, retained)
- [x] Pagination (server-side on Users; client-side on other tables)
- [x] Optimized API calls (local auth, caching, de-dupe, column projection)
- [x] Optimized rendering (memoization, skeletons, lazy routes)

Trade-offs / notes

- Aggregate caching introduces **bounded staleness** (≤15–30 s) on dashboard counts and reports after a mutation — standard for SaaS dashboards and invisible in practice. Mutating lists (Users) are **not** cached on the server and are invalidated on the client after writes, so edits appear immediately.
- Local JWKS verification keeps the **network fallback** so a legacy token or a brief JWKS outage never locks anyone out — same security, faster transport.

FILES.md — File-by-file reference

Every file in the Admin module, with: **path · purpose · MVC role · imports · exports · responsibilities · interactions · testing**. Written so a junior developer can open any file and know what it does and why.

Backend — `backend/app/admin/`

`__init__.py`

- **Role:** package entry.
- **Imports:** `from app.admin.router import router`.
- **Exports:** `router` (the aggregated `APIRouter`).
- **Responsibilities:** document the MVC module; expose the single public symbol.
- **Interactions:** imported by `app/main.py`.
- **Testing:** covered indirectly by `test_routers.py` (importing `router`).

`router.py`

- **Role:** View (aggregator).
- **Imports:** the eight sub-routers from `app.admin.routers`.
- **Exports:** `router = APIRouter(prefix="/api/v1/admin")` with all sub-routers included.
- **Responsibilities:** mount auth/dashboard/users/companies/jobs/applications/reports/settings under one prefix.
- **Interactions:** included by `main.py`; replaces the legacy monolith.
- **Testing:** `test_routers.py` mounts it in a `TestClient`.

`permissions.py`

- **Role:** cross-cutting auth dependency.
- **Imports:** `AdminRepository`, `app.auth.verify_token`.
- **Exports:** `async def get_current_admin(...) -> dict`.
- **Responsibilities:** verify the Supabase JWT (reused) then confirm the user is in `admin_profiles`; else 403.
- **Interactions:** `Depends(get_current_admin)` guards every protected router.
- **Testing:** overridden in `test_routers.py`; the live path is exercised by the smoke test.

`validators.py`

- **Role:** cross-cutting validation.
- **Imports:** `fastapi.HTTPException`.
- **Exports:** `validate_user_type`, `validate_dataset`, `validate_format`, `require_non_empty`, and the `USER_TYPES` / `REPORT_DATASETS` / `EXPORT_FORMATS` sets.
- **Responsibilities:** fail fast (400) on bad path/query input before it reaches a service/DB.
- **Interactions:** used by `UserService`, `ReportService`, the reports router.
- **Testing:** `test_validators.py` (6 tests).

repositories/ (MODEL)

`repositories/base.py`

- **Role:** Model base.
- **Imports:** `app.auth.get_supabase`.
- **Exports:** `BaseRepository` (`.client`, `.table`, `table_name`).
- **Responsibilities:** give every repository one Supabase service-role client and a `.table` query builder.

- **Interactions:** parent of all table repositories.
- **Testing:** `conftest.py` stubs `get_supabase` so subclasses construct without network.

`repositories/admin_repository.py`

- **Role:** Model — `admin_profiles`.
- **Exports:** `AdminRepository` — `list_all`, `get`, `count`, `upsert`, `update`, `delete`.
- **Responsibilities:** CRUD on admin rows.
- **Interactions:** `permissions`, `AuthService`, `SettingsService`.
- **Testing:** via service/router tests (stubbed).

`repositories/candidate_repository.py`

- **Role:** Model — `candidate_profiles`.
- **Exports:** `CandidateRepository` — `list_all`, `get`, `count`, `upsert`, `update`, `delete`.
- **Interactions:** `UserService`, `DashboardService`, `ApplicationService`, `ReportService`.

`repositories/hr_repository.py`

- **Role:** Model — `hr_profiles`.
- **Exports:** `HrRepository` — the standard set plus `list_by_company`, `update_company_fields` (fan-out company_* edits).
- **Interactions:** `UserService`, `CompanyService`, `DashboardService`, `ReportService`.

`repositories/job_pool_repository.py`

- **Role:** Model — `job_pools`.
- **Exports:** `JobPoolRepository` — `list_all`, `list_with_recruiter` (joins `hr_profiles`), `get`, `list_by_hr`, `count(active_only)`, `update`, `delete`.
- **Interactions:** `JobService`, `CompanyService`, `DashboardService`, `ApplicationService`, `ReportService`.

`repositories/interview_repository.py`

- **Role:** Model — `interview_sessions`.
- **Exports:** `InterviewRepository` — `list_all`, `list_for_session`, `list_for_candidate`, `count(status)`, `delete_session`.
- **Interactions:** `ApplicationService`, `DashboardService`.

`repositories/summary_repository.py`

- **Role:** Model — `Candidate_summaries` (mixed-case, quoted exactly).
- **Exports:** `SummaryRepository` — `list_all`, `get_for_candidate`, `count`.
- **Interactions:** `ApplicationService` (AI summary), `DashboardService` (count).

`repositories/auth_repository.py`

- **Role:** Model — Supabase Auth (GoTrue) admin API (no DB table).
- **Imports:** `httpx`, `app.auth.get_supabase`, config.
- **Exports:** `AuthRepository` — `create_user`, `delete_user`, `set_password`, `sign_in`, `ban_user`, `unban_user`, `get_auth_user`, `is_disabled`, `disabled_map`.
- **Responsibilities:** account lifecycle + **disable-via-ban** + reading `banned_until`.
- **Interactions:** `UserService`, `AuthService`, `SettingsService`.
- **Testing:** exercised live (create/ban happen against GoTrue); not stubbed in unit tests.

schemas/ (CONTRACTS)

- `schemas/common.py` — `MessageOut` (generic `{message, ok}`).
- `schemas/auth_schemas.py` — `AdminLogin`, `AdminRegister`, `AdminOut`, `AdminToken`.
- `schemas/dashboard_schemas.py` — `AdminStats` (KPI counts), `ActivityItem`.

- `schemas/user_schemas.py` — `CandidateCreate/Update`, `RecruiterCreate/Update` (only real columns; updates all-optional/PATCH).
- `schemas/company_schemas.py` — `CompanyUpdate` (company_* fields).
- `schemas/job_schemas.py` — `JobUpdate` (editable job_pools columns, arrays → text[]).
- `schemas/settings_schemas.py` — `AdminCreate`, `AdminProfileUpdate`, `PasswordChange`.
- **Role:** request/response contracts. **Exports:** the pydantic models (re-exported from `schemas/__init__.py`). **Interactions:** routers (request bodies + `response_model`). **Testing:** validated implicitly by router tests.

services/ (CONTROLLER)

`services/auth_service.py`

- **Exports:** `AuthService` — `login`, `register`, `me`.
- **Responsibilities:** sign in via Supabase then gate on `admin_profiles`; bootstrap first admin (guarded by `ADMIN_SETUP_TOKEN`); roll back the auth user if the profile insert fails.
- **Interactions:** `AdminRepository`, `AuthRepository`; used by `auth_router`.

`services/dashboard_service.py`

- **Exports:** `DashboardService` — `stats`, `charts`, `activity`.
- **Responsibilities:** KPI counts; 6-month signup growth; application-status donut; pools-by-state bar; top companies; merged recent-activity feed.
- **Interactions:** candidate/hr/job/interview/summary repos + `ApplicationService`.

`services/user_service.py`

- **Exports:** `UserService` — `list_users`, `get_user`, `create_candidate`, `create_recruiter`, `update_user`, `set_disabled`, `delete_user`.
- **Responsibilities:** unify candidates + recruiters; overlay disabled status; normalise rows; enforce `validate_user_type`; disable-via-ban; rollback on failed create.
- **Interactions:** candidate/hr repos + `AuthRepository`. **Excludes admins by construction.**

`services/company_service.py`

- **Exports:** `CompanyService` — `list_companies`, `get_company`, `update_company`.
- **Responsibilities:** aggregate `hr_profiles` by `company_name`; count recruiters/jobs; first-non-null company fields; attach pools on detail; fan-out company_* edits.
- **Interactions:** `HrRepository`, `JobPoolRepository`.
- **Testing:** `test_company_and_report.py`.

`services/job_service.py`

- **Exports:** `JobService` — `list_jobs`, `get_job`, `update_job`, `delete_job` + `_flatten` helper.
- **Responsibilities:** flatten the joined recruiter/company; status filter (active/inactive/archived); search; PATCH changed fields.
- **Interactions:** `JobPoolRepository`.

`services/application_service.py`

- **Exports:** `ApplicationService` — `list_applications`, `get_application`.
- **Responsibilities:** group `interview_sessions` rows by `session_id` into applications; compute progress/status; join candidate/pool/AI summary; build transcript.
- **Interactions:** interview/candidate/job/summary repos.
- **Testing:** `test_application_service.py`.

`services/report_service.py`

- **Exports:** `ReportService` — `summary`, `build_rows`, `to_csv`, `to_xlsx`; helpers `_cell`, `JobServiceRows`.
- **Responsibilities:** analytics totals/rates; per-dataset column sets; CSV (UTF-8 BOM) and styled XLSX, both in-memory.

- **Interactions:** candidate/hr/job repos + Application/Company services.
- **Testing:** `test_company_and_report.py` (CSV/XLSX/cell).

`services/settings_service.py`

- **Exports:** `SettingsService` — `list_admins`, `create_admin`, `delete_admin`, `update_profile`, `change_password`, `api_keys`.
- **Responsibilities:** admin-user management with guards (no self-delete, no last-admin delete); strip `password` from output; password change via GoTrue; api-keys placeholder.
- **Interactions:** `AdminRepository`, `AuthRepository`.

routers/ (VIEW)

Each is a thin `APIRouter`; every protected endpoint uses `Depends(get_current_admin)`.

- `routers/auth_router.py` — `/login`, `/register`, `/me`, `/logout`, `/health` → `AuthService`.
- `routers/dashboard_router.py` — `/dashboard/stats|charts|activity` → `DashboardService`.
- `routers/users_router.py` — `GET/POST/PUT/DELETE /users...`, `.../disable`, `.../enable` → `UserService`.
- `routers/companies_router.py` — `GET /companies`, `GET/PUT /companies/{key}` → `CompanyService`.
- `routers/jobs_router.py` — `GET /jobs`, `GET/PATCH/DELETE /jobs/{id}` → `JobService`.
- `routers/applications_router.py` — `GET /applications`, `GET /applications/{session_id}` → `ApplicationService`.
- `routers/reports_router.py` — `GET /reports/summary`, `GET /reports/export` (CSV/XLSX `Response`) → `ReportService` + validators.
- `routers/settings_router.py` — `/settings/admins`, `/settings/profile`, `/settings/security/password`, `/settings/api-keys` → `SettingsService`.
- **Testing:** wiring smoke-tested in `test_routers.py`; full surface exercised live.

tests/

- `conftest.py` — autouse fixture stubbing `get_supabase` (3 import sites) so repos build offline.
- `test_validators.py`, `test_application_service.py`, `test_company_and_report.py`, `test_routers.py` — see testing.md. **23 tests, all passing.**

Frontend — `frontend/src/admin/`

`AdminApp.tsx`

- **Role:** View (module entry).
- **Imports:** providers, `RequireAdmin`, `AdminLayout`, all pages, `Toaster`.
- **Exports:** default `AdminApp`.
- **Responsibilities:** wrap the module in `AdminAuthProvider` + `ConfirmProvider` + toaster; declare the route table (login public; rest guarded inside the layout).
- **Interactions:** mounted by `App.tsx` at `/admin/*`.

`types.ts`

- **Role:** Model (contracts). **Exports:** `Admin`, `UserRow`, `AdminStats`, `DashboardCharts`, `ActivityItem`, `Company`, `Job`, `ApplicationRow`, `ApplicationDetail`, `ReportSummary`, `UserType`.
- **Responsibilities:** one source of truth for API shapes. **Interactions:** services + pages.

context/ `AdminAuthContext.tsx`

- **Role:** Controller. **Exports:** `AdminAuthProvider`, `useAdminAuth`.
- **Responsibilities:** hold current admin; `login/logout/refresh/setAdmin`; validate token on mount; keep session on transient errors, drop only on real auth failure.
- **Interactions:** `auth.service`, `client.isAuthError`; consumed by RequireAdmin/Topbar/LoginPage/Settings.

hooks/

- `useAsync.ts` — Controller. `useAsync(fn, deps)` → `{data, loading, error, reload}`; maps auth errors to a friendly message. Used by every list/detail page.
- `useDebounce.ts` — Controller. Debounce a value (search boxes).
- `useToast.ts` — Controller. `toast.success/error/...` over react-hot-toast.

lib/ `format.ts`

- **Role:** helper. **Exports:** `formatDate`, `formatDateTime`, `timeAgo`, `titleCase`, `displayValue`. Used across pages/drawers for consistent rendering.

services/ (MODEL)

- `client.ts` — the `http` wrapper (`get/post/put/patch/del/download`), token storage (`getAdminToken` / `setAdminToken` / `clearAdminToken`), `AdminApiError`, `isAuthError`. **Every** request goes through here.
- `auth.service.ts` — `login`, `getCurrentAdmin`, `logout`, `hasToken`.
- `dashboard.service.ts` — `getStats`, `getCharts`, `getActivity`.
- `users.service.ts` — `listUsers`, `getUser`, `createCandidate`, `createRecruiter`, `updateUser`, `disableUser`, `enableUser`, `deleteUser`.
- `companies.service.ts` — `listCompanies`, `getCompany`, `updateCompany`.
- `jobs.service.ts` — `listJobs`, `getJob`, `updateJob`, `deleteJob`.
- `applications.service.ts` — `listApplications`, `getApplication`.
- `reports.service.ts` — `getSummary`, `exportDataset` (CSV/XLSX download).
- `settings.service.ts` — `listAdmins`, `createAdmin`, `deleteAdmin`, `updateProfile`, `changePassword`, `getApiKeys`.

components/ui/ (design system)

- `Button.tsx` — variants (primary/secondary/outline/ghost/destructive) + sizes + `loading` spinner.
- `form.tsx` — `Label`, `Input`, `Textarea`, `Select`, `Field` (label+control+hint/error).
- `primitives.tsx` — `Card`, `Badge`, `StatusBadge`, `Spinner`, `Skeleton`, `LoadingState`, `EmptyState`, `ErrorState`, `PageHeader`, `Avatar`.
- `SearchInput.tsx` — icon + clear-button search box.
- `Drawer.tsx` — right-hand slide-over (Escape/backdrop close, scroll-lock) for detail/edit.
- `Modal.tsx` — centered dialog (create forms).
- `ActionsMenu.tsx` — three-dot dropdown (portal-positioned; danger/disabled items).
- `ConfirmDialog.tsx` — `ConfirmProvider` + `useConfirm()` imperative confirm for destructive actions.
- `KpiCard.tsx` — dashboard stat card (icon, value, hint, loading skeleton).
- `Table.tsx` — generic `DataTable<T>` (columns, `rowKey`, loading/empty/error built in) — the reason there are **no duplicate-key warnings**.
- `Tabs.tsx` — underline tabs (Settings sections).
- `index.ts` — barrel re-exporting all of the above.

components/charts/ `Charts.tsx`

- **Role:** View. **Exports:** `GrowthAreaChart`, `StatusDonut`, `SimpleBarChart` (recharts, xQuesty palette, responsive). Used by Dashboard + Reports.

components/layout/

- `RequireAdmin.tsx` — guard: no token → redirect; validating → loader; validated → children.
- `Sidebar.tsx` — xQuesty-branded nav (Dashboard/Users/Companies/Jobs/Applications/Reports). **No Settings link** (spec).
- `Topbar.tsx` — mobile menu button + **profile dropdown** (My Profile, Settings, Sign out).
- `AdminLayout.tsx` — responsive shell: sidebar + topbar + `<Outlet/>`; off-canvas sidebar on mobile.

components/users/

- `CreateUserModal.tsx` — role-switching create form (candidate/recruiter) → `users.service`.
- `UserDetailDrawer.tsx` — large view+edit drawer; fetches full profile; field groups per type; PATCHes only changed fields.

components/companies/ `CompanyDetailDrawer.tsx`

- Company overview + editable company_* fields + member recruiters + pools.

components/jobs/ `JobDetailDrawer.tsx`

- Job overview/requirements + management actions (activate/deactivate, archive/unarchive, edit, delete-handled-on-page).

components/applications/ `ApplicationDetailDrawer.tsx`

- Candidate/job context, progress, AI summary (Candidate_summaries), full Q&A transcript (read-only).

pages/ (VIEW)

- `LoginPage.tsx` — branded login; redirects authenticated admins to the dashboard.
- `DashboardPage.tsx` — KPI cards, growth/donut/bar charts, activity feed, quick actions.
- `UsersPage.tsx` — unified table; search/type filter; 3-dot actions (view/edit/disable/enable/delete); create modal; detail drawer.
- `CompaniesPage.tsx` — searchable company card grid → detail drawer.
- `JobsPage.tsx` — table; search/status filter; actions; detail drawer.
- `ApplicationsPage.tsx` — table with progress bars; search/status filter; transcript drawer.
- `ReportsPage.tsx` — analytics totals, completion gauge, top-companies chart, CSV/Excel export grid.
- `SettingsPage.tsx` — tabbed (My Profile, Admin Users, Security, API Keys placeholder); URL-encoded section; create/delete admins with guards.

Performance layer (added in the optimization pass)

These files were introduced/changed to make the dashboard feel instant. See PERFORMANCE_AUDIT.md and SPRINT_REPORT.md.

Backend

`app/admin/cache.py` *(new)*

- **Role:** cross-cutting (performance). **Exports:** `TTLCache`, `ttl_cached`.
- **Responsibilities:** dependency-free, thread-safe time-to-live cache and a memoize decorator for read-only service aggregates.
- **Interactions:** used by `security.py`, `auth_repository.py`, and the dashboard/ application/report services.
- **Testing:** exercised indirectly; `tests/conftest.py` clears these caches between tests so memoized results don't leak across cases.

`app/admin/security.py` ***(new)***

- **Role:** cross-cutting (auth performance). **Exports:** `verify_admin_token`, `get_admin_profile`, `invalidate_admin`.
- **Responsibilities:** verify the JWT **locally** via Supabase JWKS (ES256), with a cached result and a network fallback; cache the `admin_profiles` lookup. This removes the ~376 ms-per-request GoTrue round trip that was the root cause.
- **Interactions:** used by `permissions.get_current_admin`.

Changed backend files

- `permissions.py` — uses `security.verify_admin_token` + `get_admin_profile`.
- `repositories/base.py` — single shared service-role client (`get_shared_supabase`).
- `repositories/auth_repository.py` — `disabled_map` cached (30 s) + invalidated on ban/unban.
- `repositories/candidate_repository.py`, `hr_repository.py` — `list_basic` projection.
- `services/dashboard_service.py` — `stats` / `charts` / `activity` `@ttl_cached(15)`.
- `services/application_service.py` — `list_applications` `@ttl_cached(15)`.
- `services/report_service.py` — `summary` `@ttl_cached(20)`.
- `services/user_service.py` + `routers/users_router.py` — server-side pagination.

Frontend

`src/admin/services/cache.ts` ***(new)***

- **Role:** Model (performance). **Exports:** `getCached`, `isFresh`, `setCached`, `invalidate`, `dedupe`, `clearCache`, `FRESH_MS`.
- **Responsibilities:** in-memory stale-while-revalidate store + in-flight request de-dupe; `invalidate(prefix)` clears affected lists after mutations.

`src/admin/hooks/useQuery.ts` ***(new)***

- **Role:** Controller. **Exports:** `useQuery(key, fetcher)`.
- **Responsibilities:** cached fetching — instant render of cached data, background revalidation, `loading` only on first fetch (drives skeletons), `reload()` to force.
- **Interactions:** used by Dashboard, Users, Companies, Jobs, Applications, Reports.

`src/admin/components/ui/Pagination.tsx` ***(new)***

- **Role:** View. **Exports:** `Pagination`. "Showing X–Y of N" + prev/next. Used by the Users page (server-side) and the `DataTable` client-side mode.

Changed frontend files

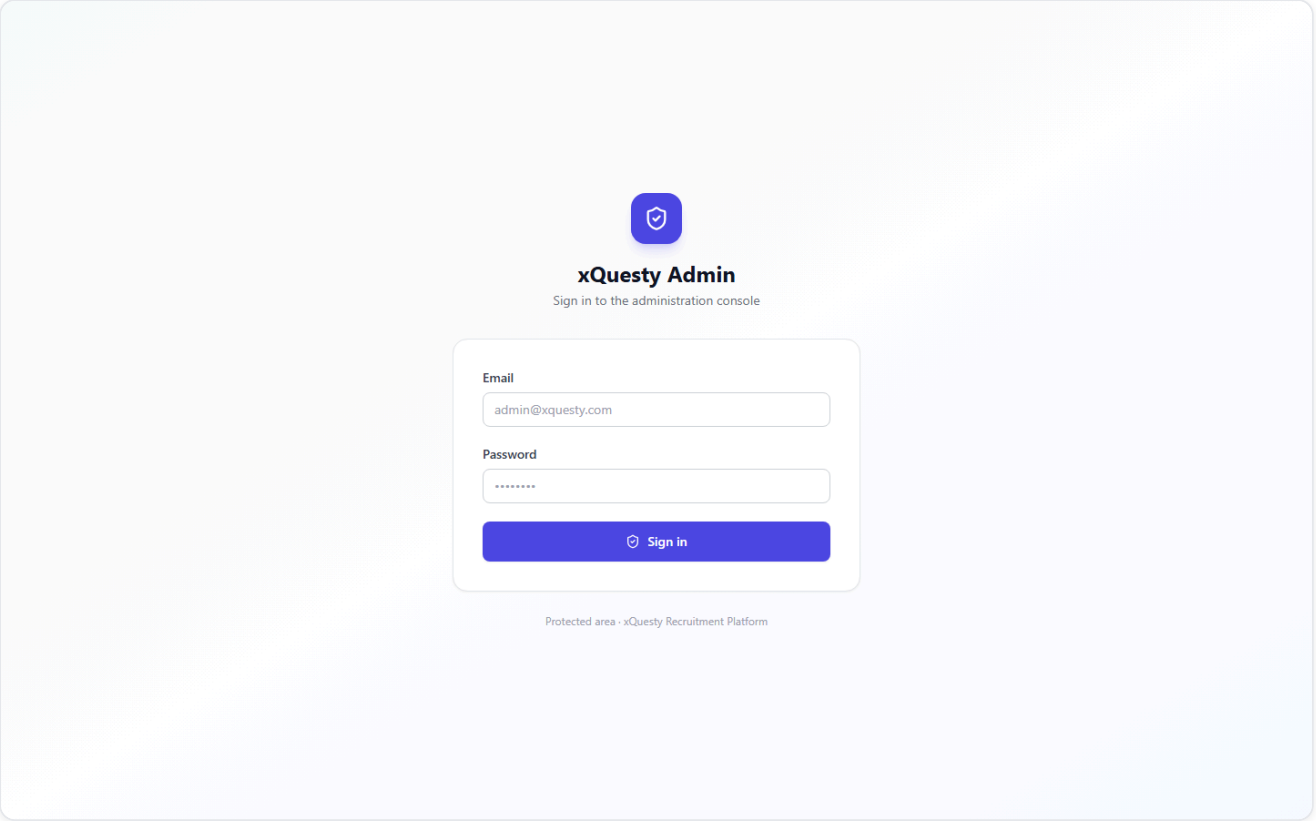
- `components/ui/Table.tsx` — skeleton rows on first load; optional `pageSize` client-side pagination.
- `components/ui/primitives.tsx` — `ChartSkeleton`; `Skeleton` accepts `style`.
- `services/users.service.ts` — paginated `listUsers` + `invalidateUsers`.
- `types.ts` — `Paginated<T>` envelope.
- `pages/*` — migrated `useAsync` → `useQuery`; memoized columns; pagination; cache invalidation on mutation.
- `AdminApp.tsx` — routes lazy-loaded (`React.lazy` + `Suspense`).

Host-app touch points (edited, not part of the module)

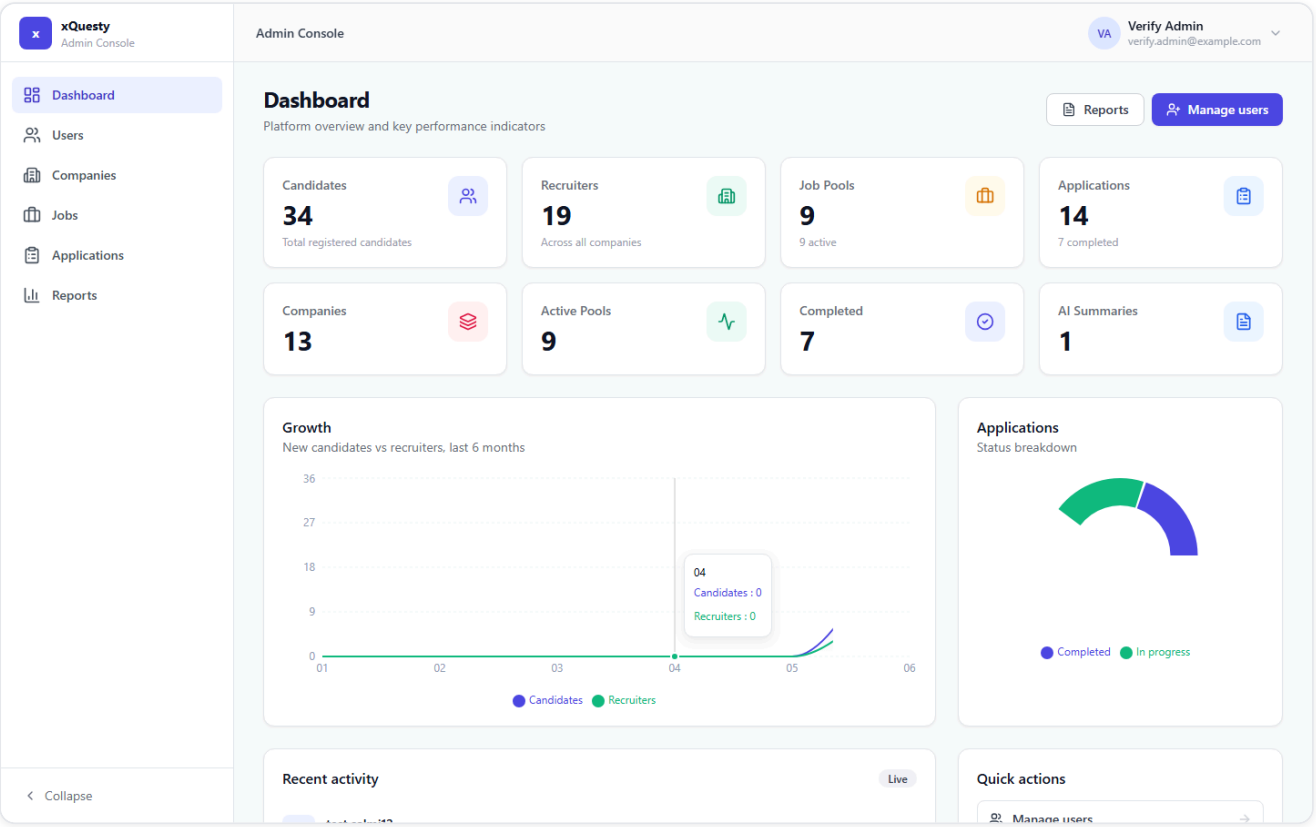
- `frontend/src/App.tsx` — mounts `<Route path="/admin/*" element={<AdminApp/>} />`.
- `backend/app/main.py` — `include_router(admin_module)`.
- `backend/requirements.txt` — added `openpyxl`, `playwright`, `pytest`.

Screenshots

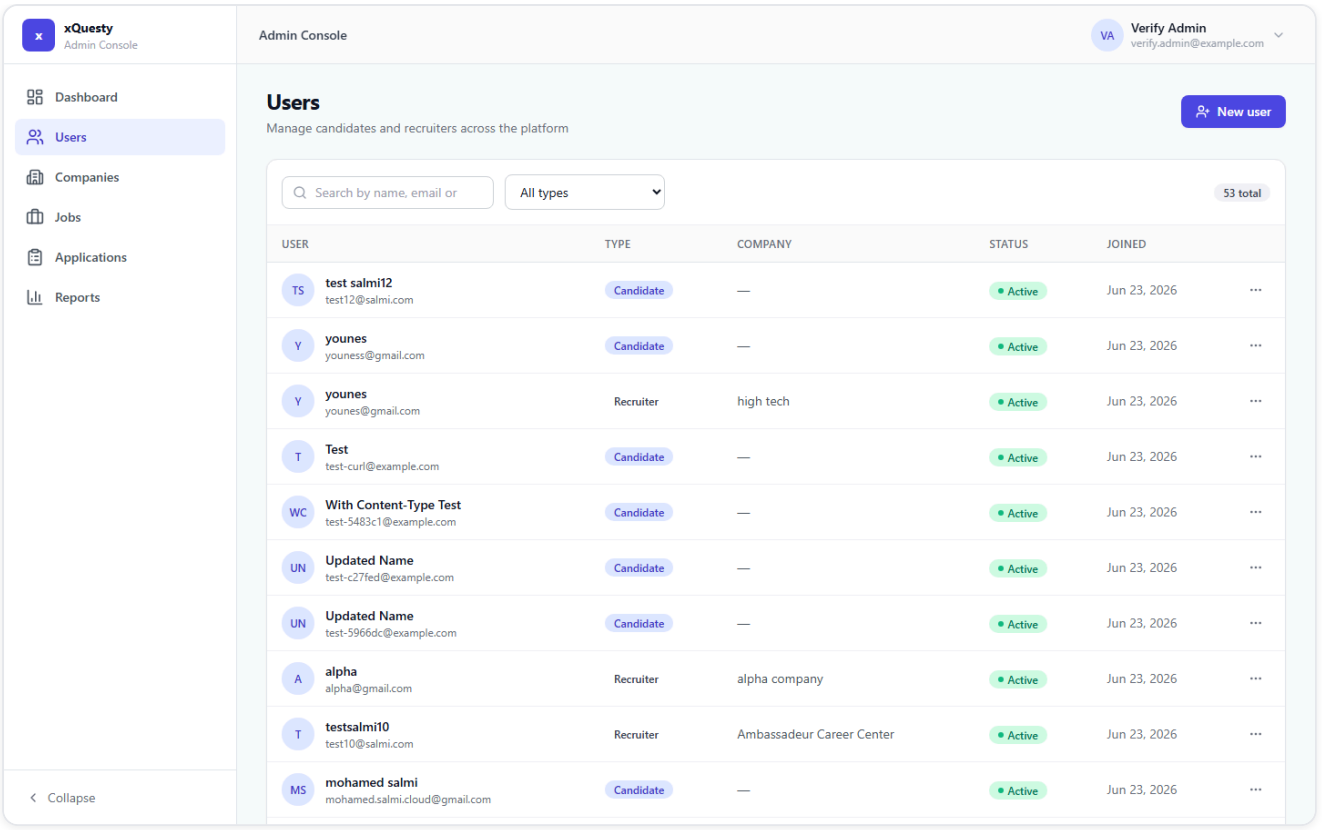
Captured from the live application driving real data (Playwright). Files in `screenshots/`.



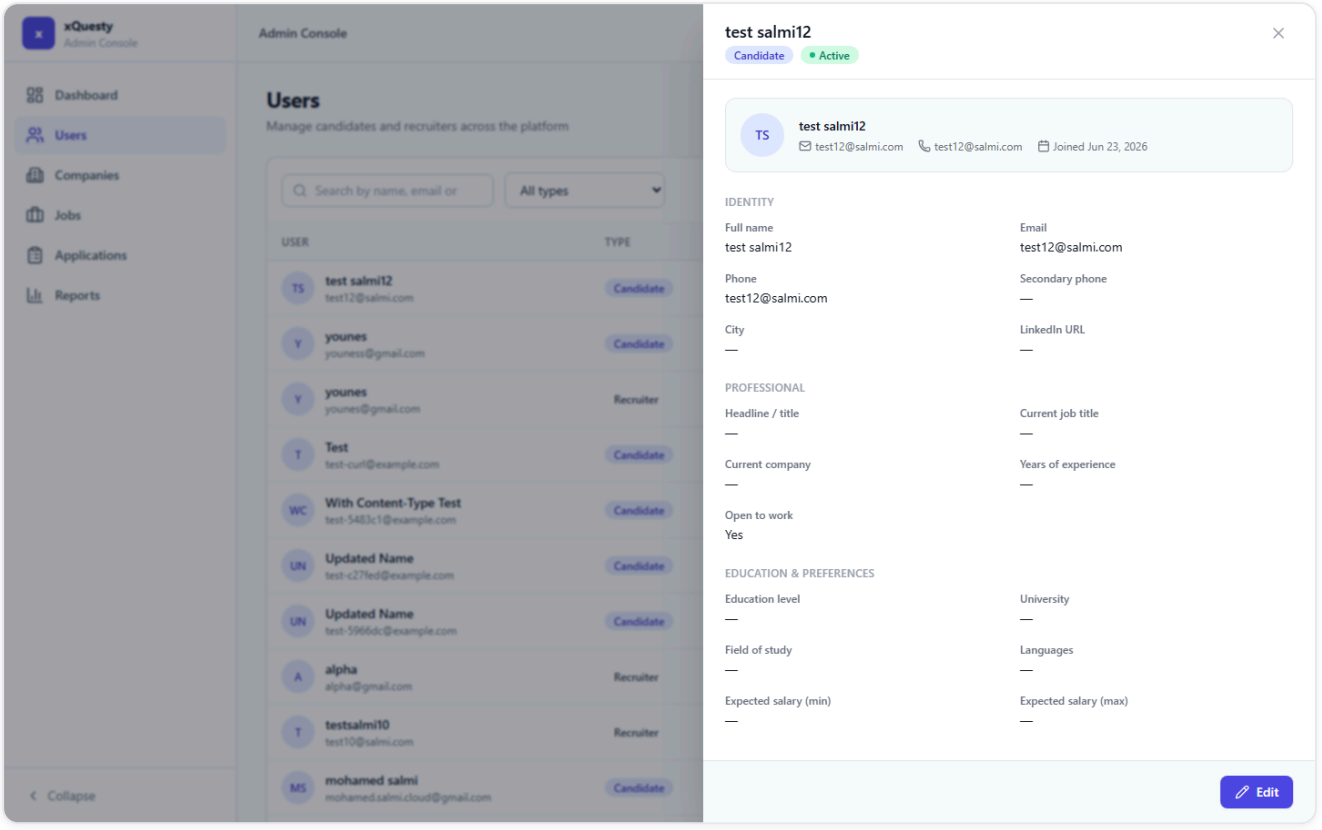
Admin login — branded, accessible sign-in.



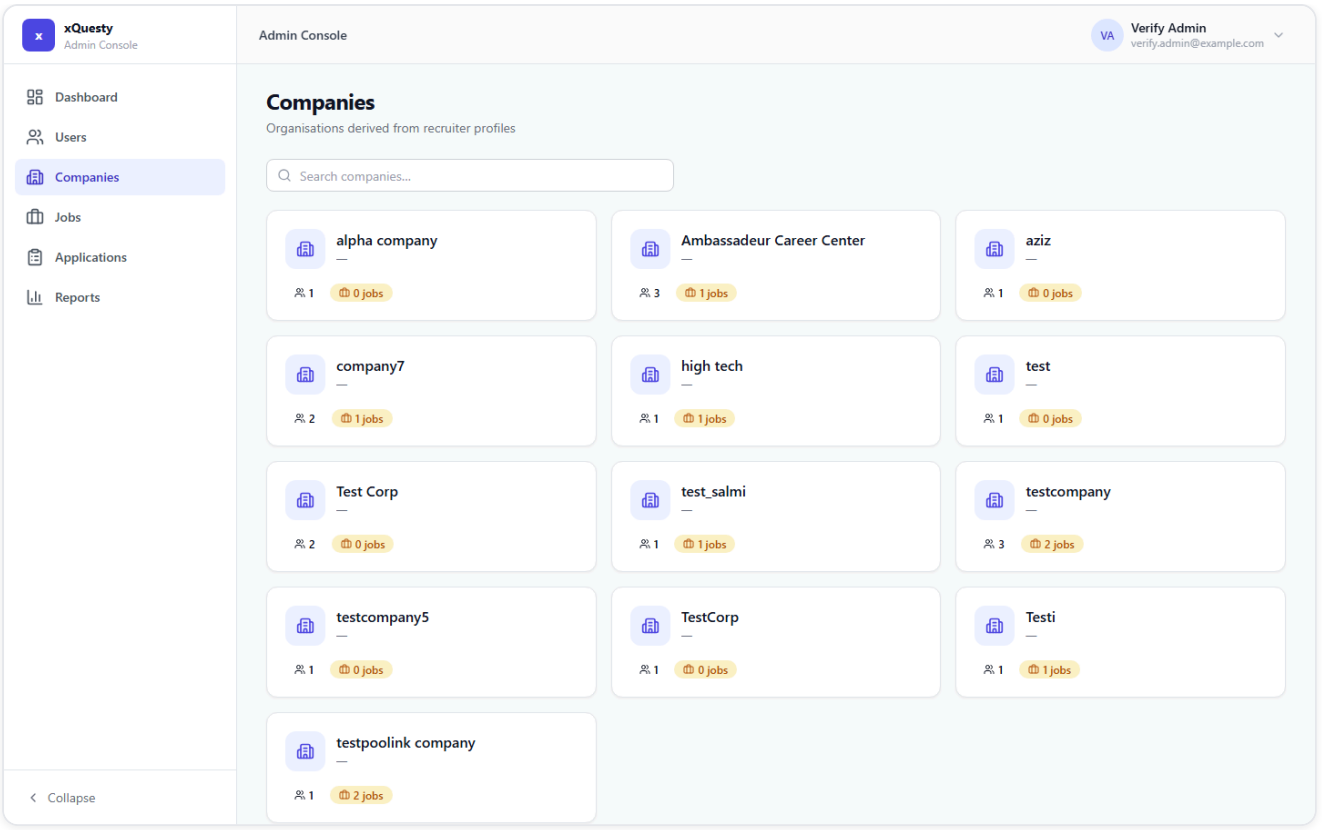
Dashboard — KPI cards, growth & status charts, activity, quick actions.



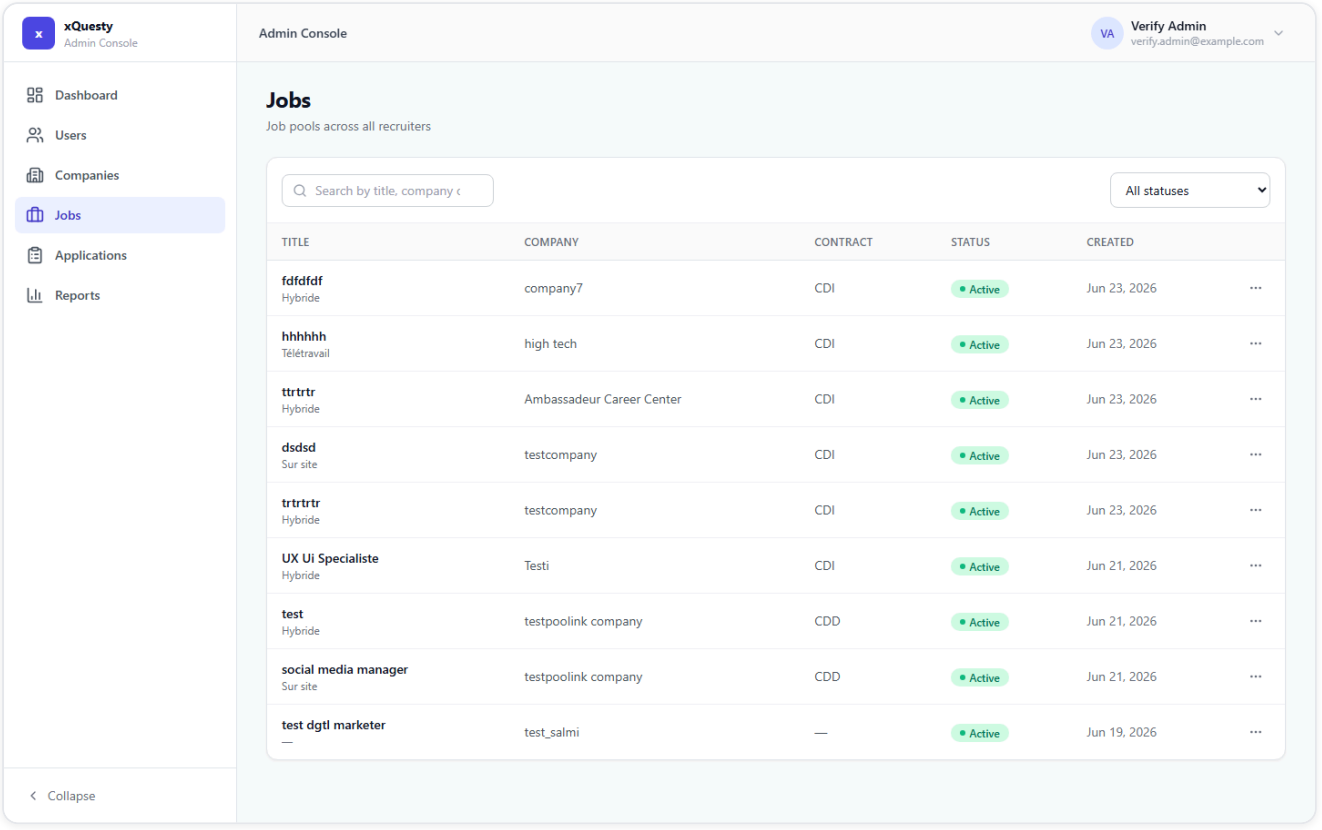
Users — unified candidates + recruiters with search, filters and 3-dot actions.



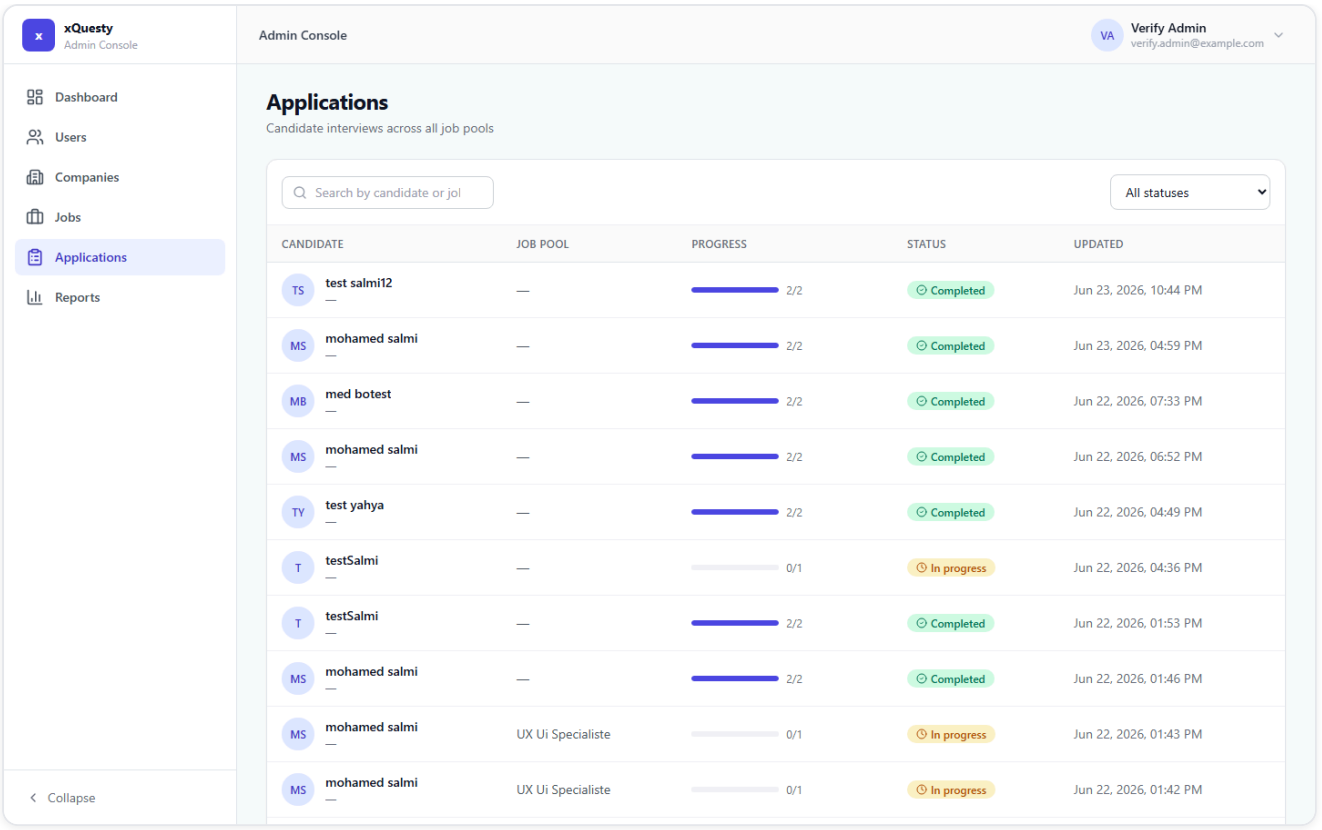
User detail drawer — large view/edit of every schema field.



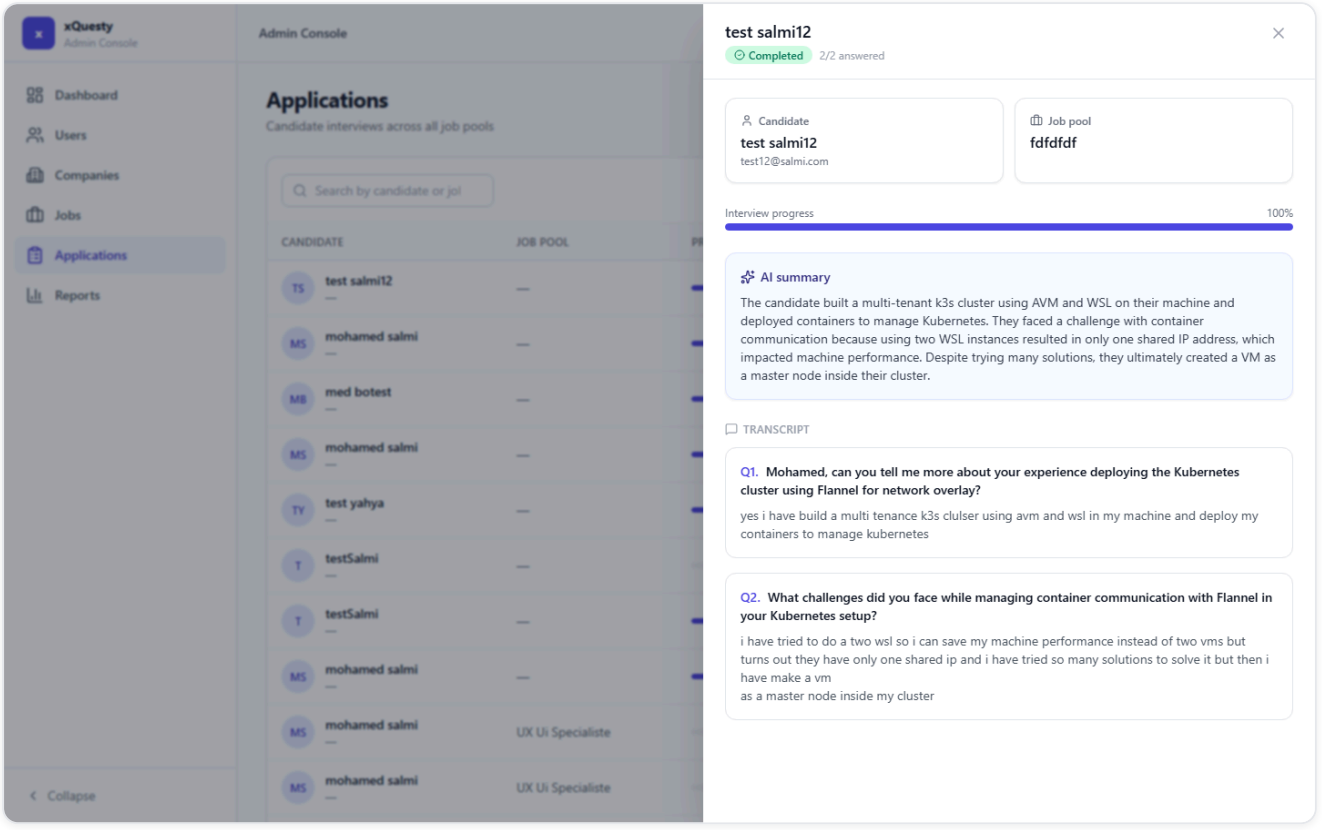
Companies — derived from hr_profiles, searchable card grid.



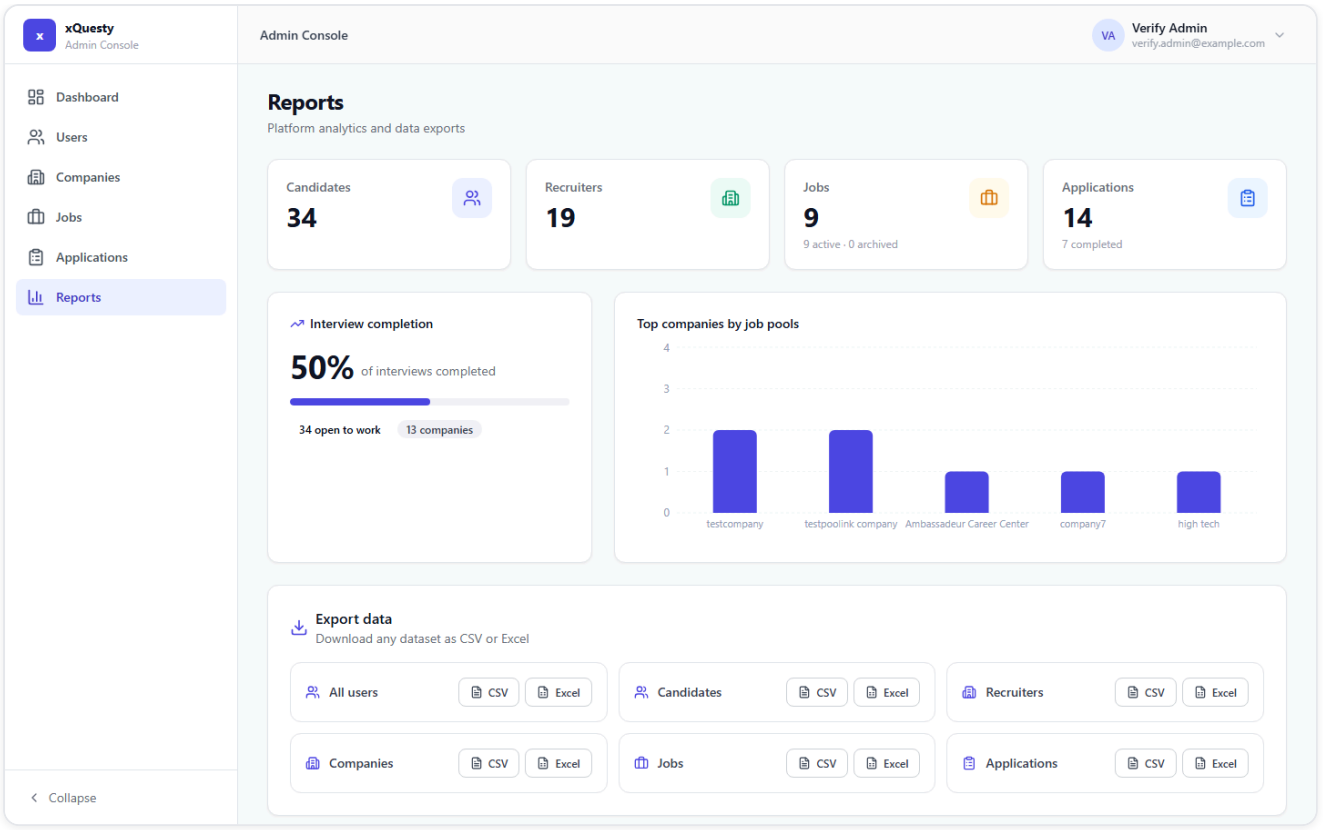
Jobs — job_pools with status filters and management actions.



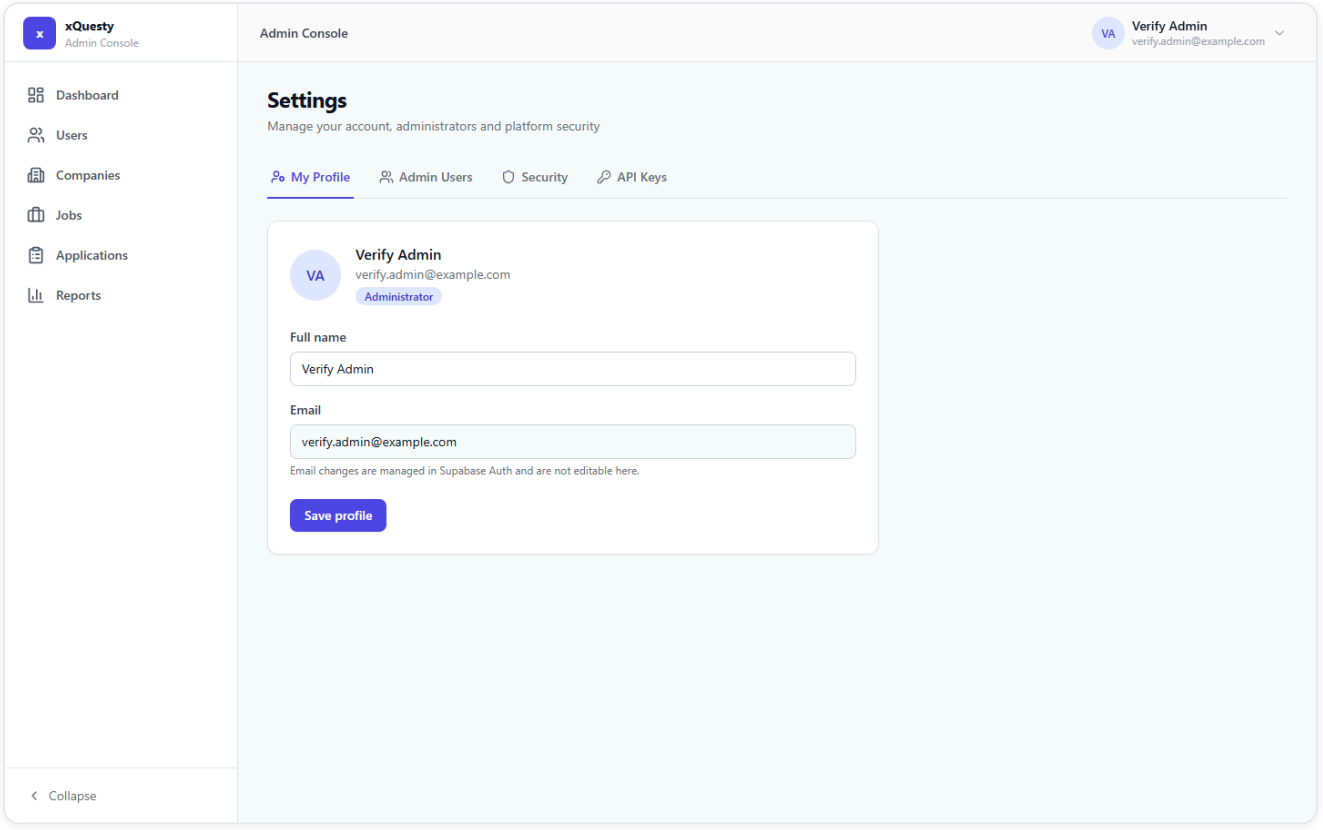
Applications — interview_sessions with progress and status.



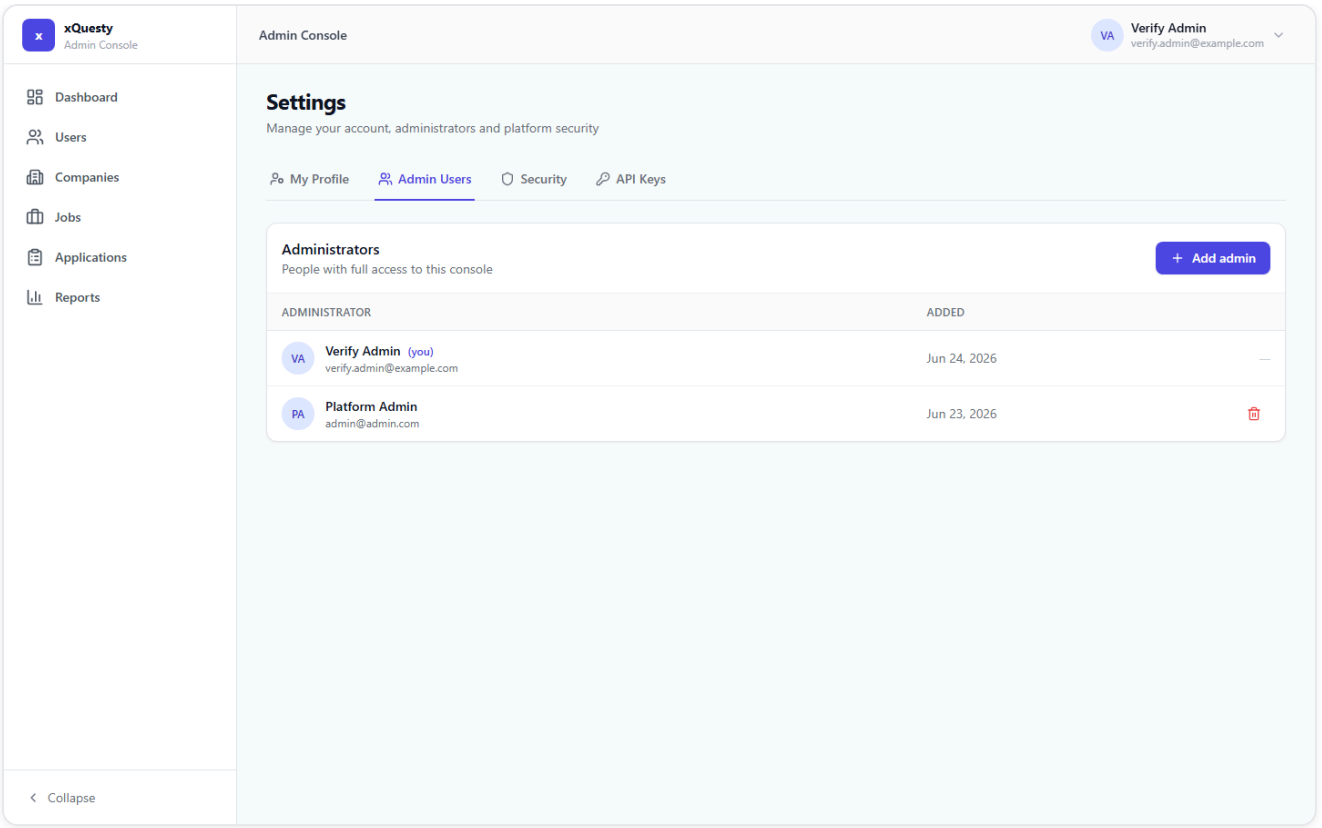
Application detail — AI summary (Candidate_summaries) + transcript.



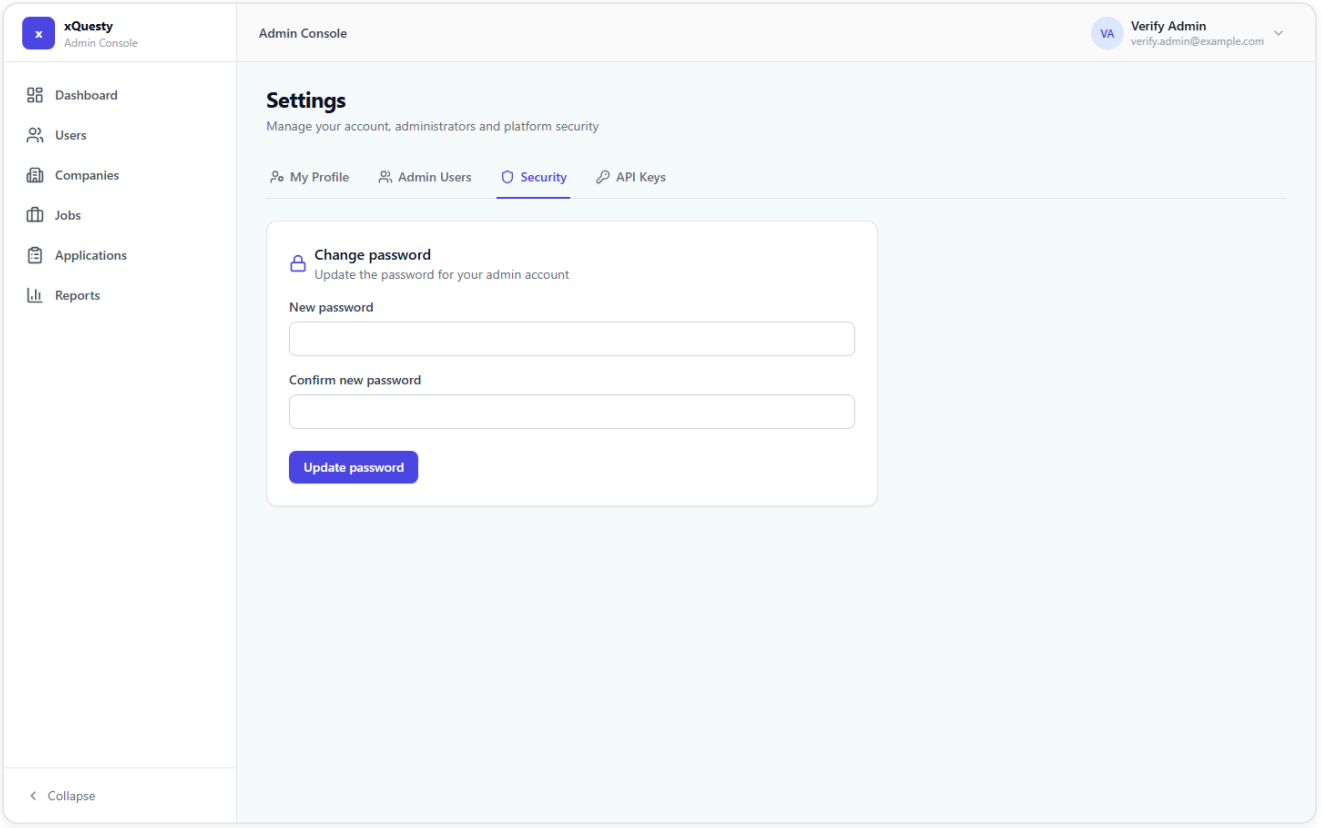
Reports — analytics, completion gauge, top companies, CSV/Excel export.



Settings → My Profile (reached via the profile dropdown).



Settings → Admin Users — the only admin-management surface.



Settings → Security — change password.

VA

Dashboard

Platform overview and key performance indicators

Reports

Manage users

Candidates

34

Total registered candidates

Recruiters

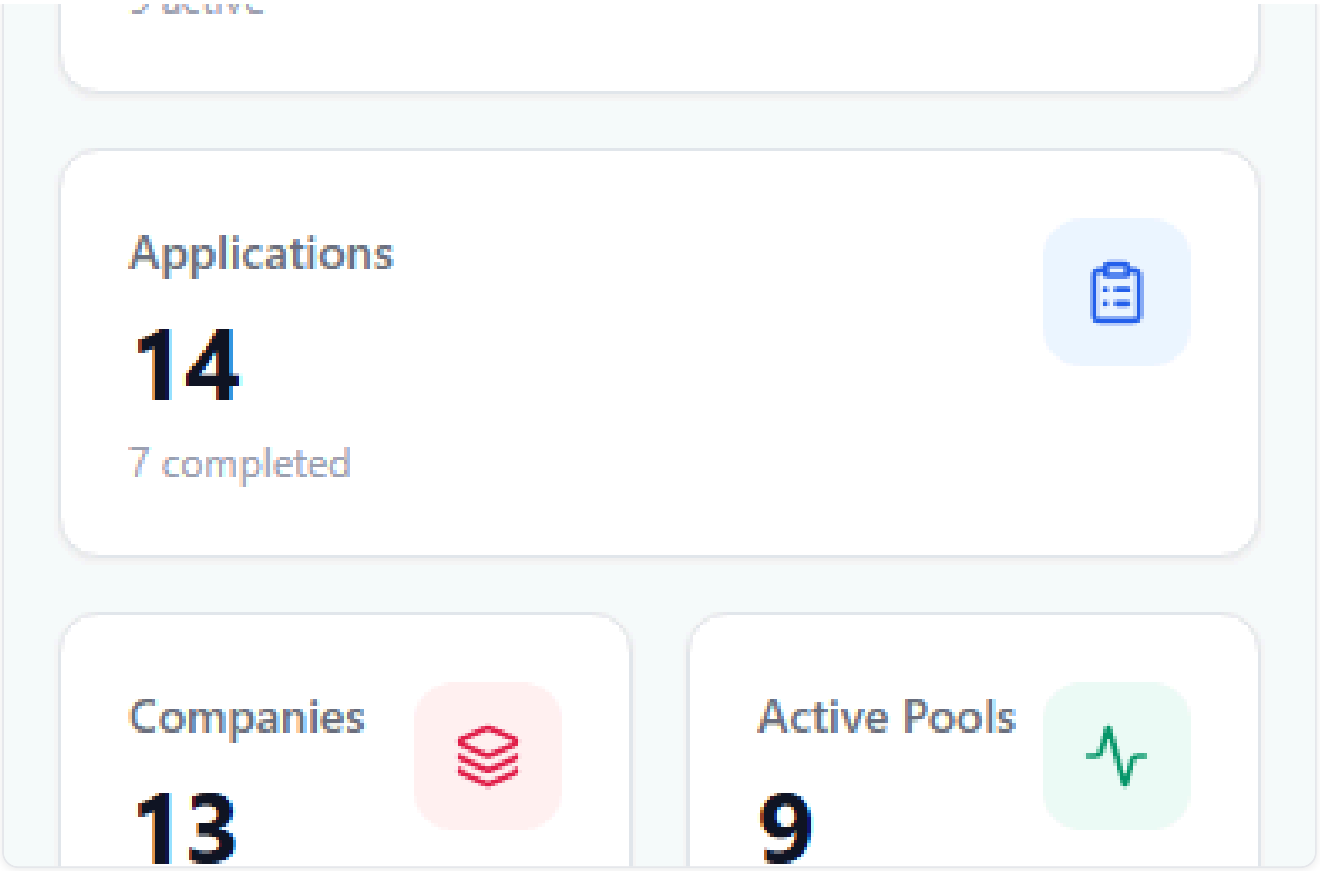
19

Across all companies

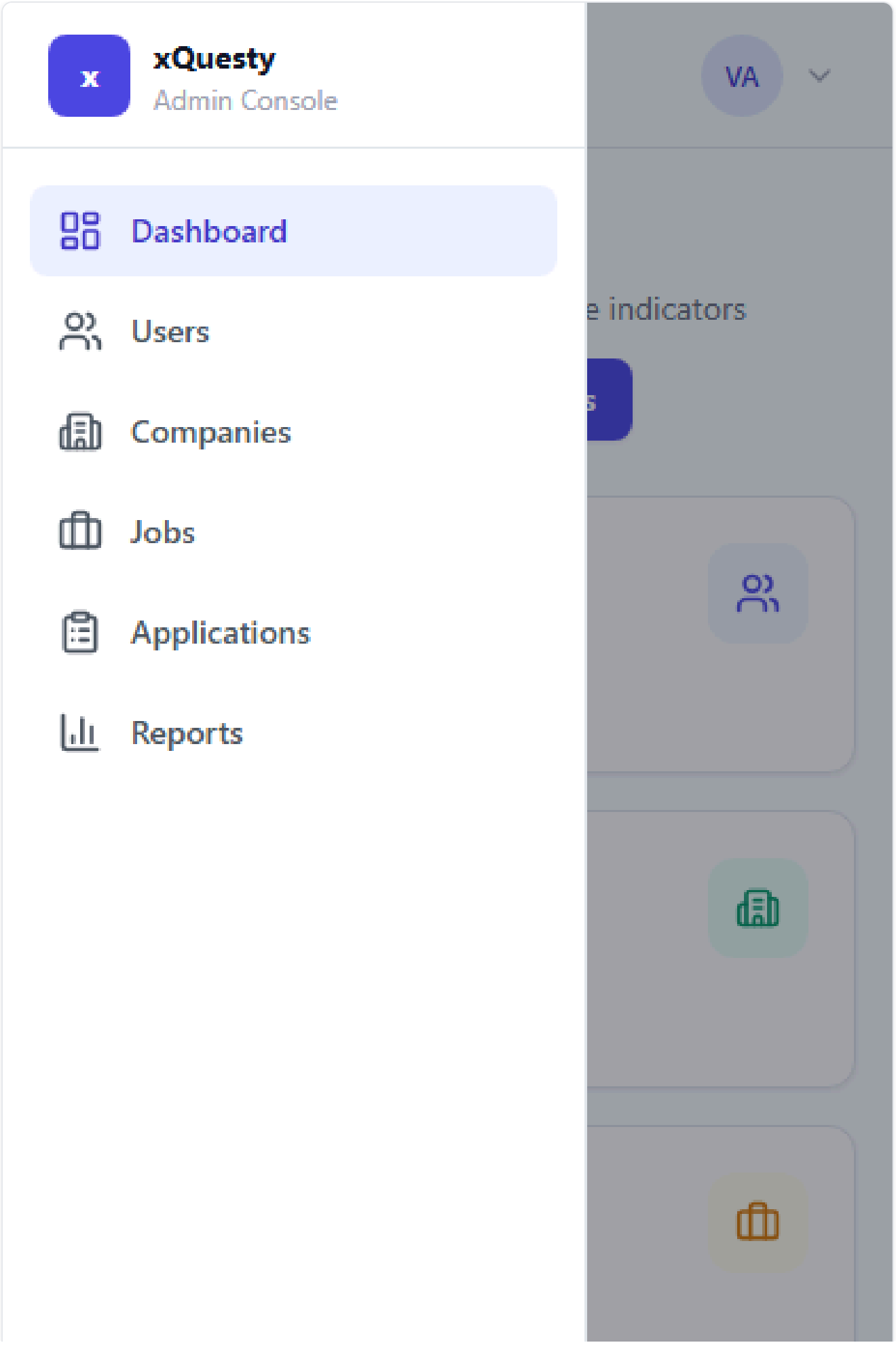
Job Pools

9

9 active



Responsive — dashboard at 390px (mobile).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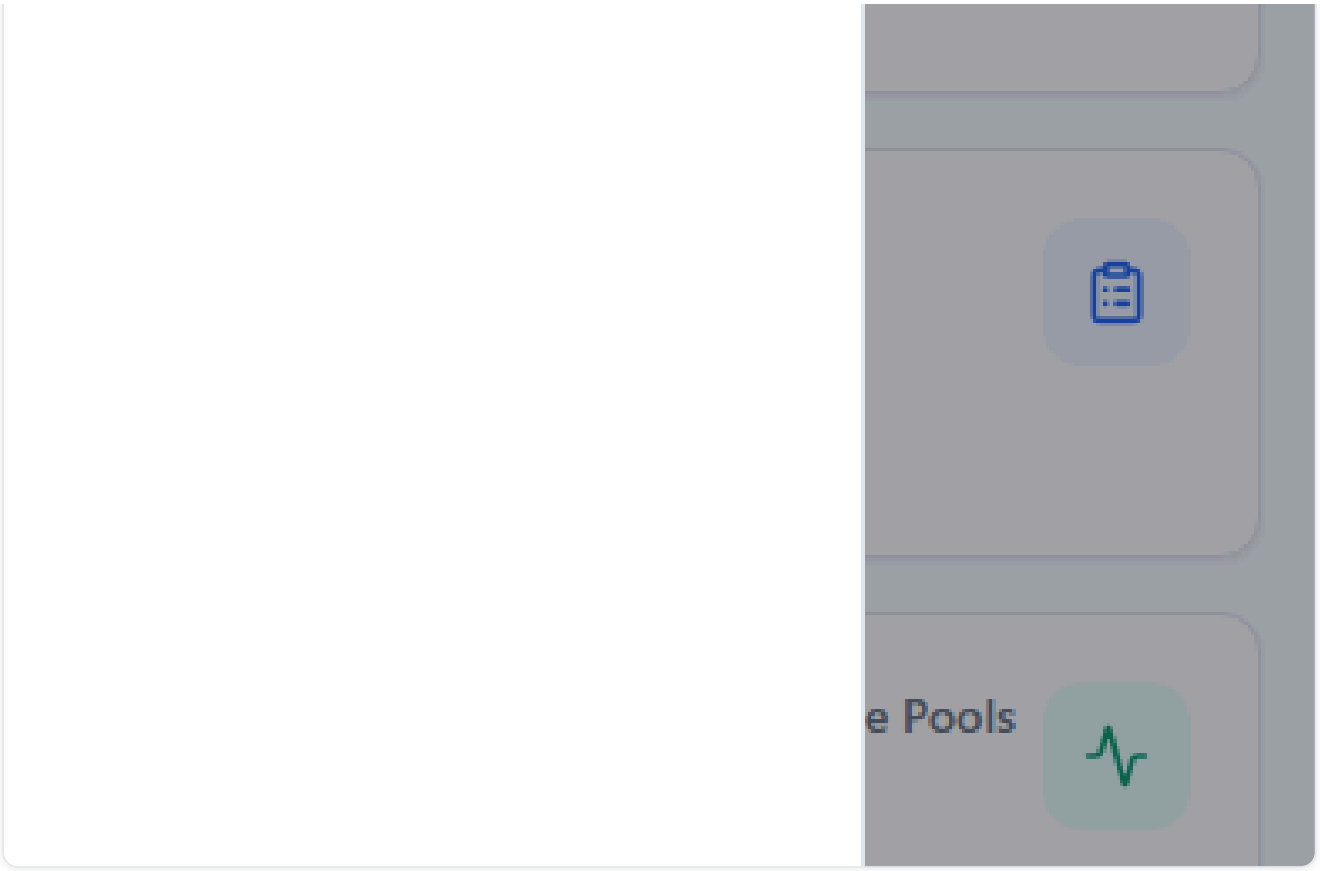
Admin Console
VA
Dashboard
Users
Companies
Jobs
Applications
Reports
e indicators



Responsive — off-canvas navigation drawer on mobile.